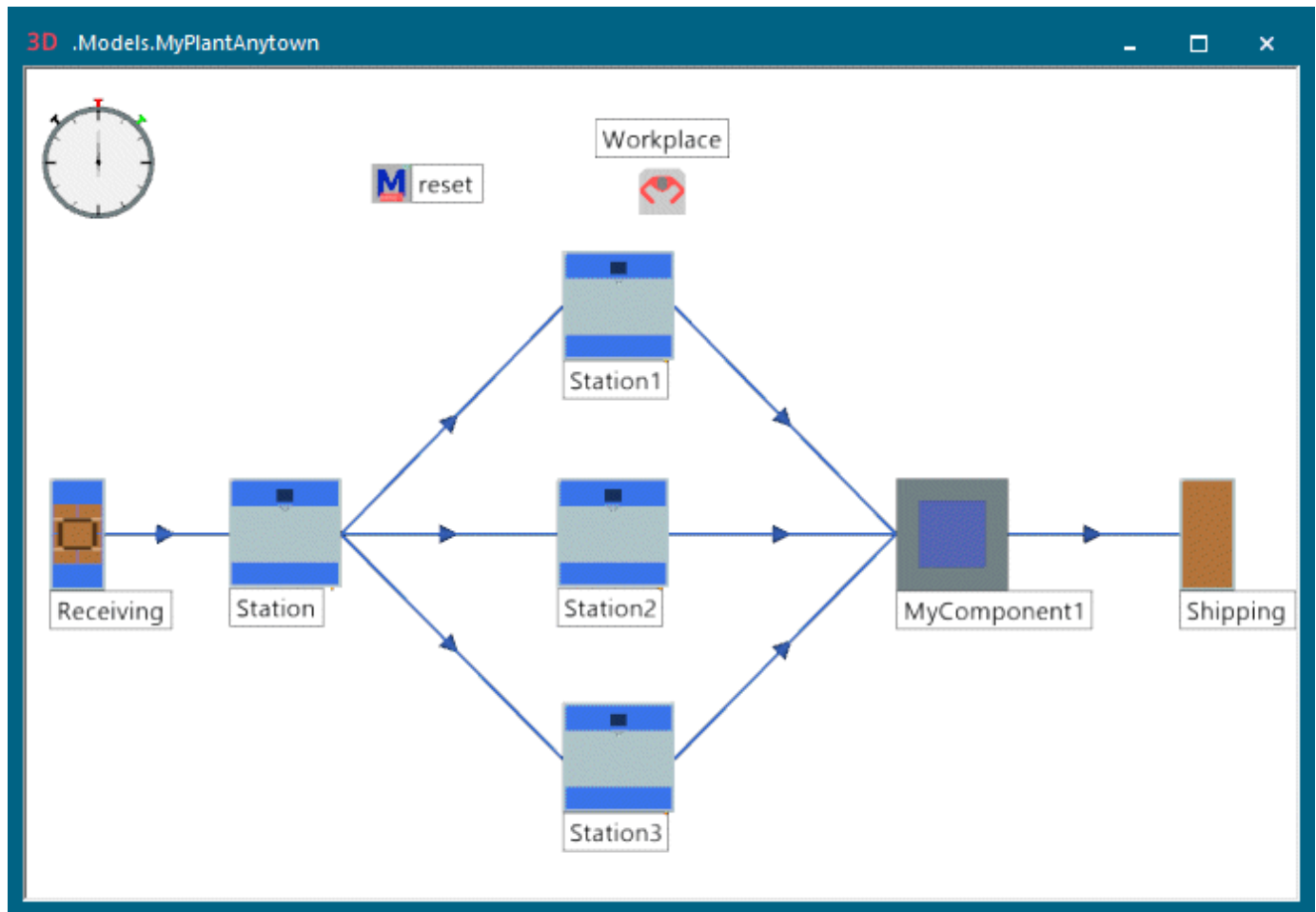


Back to [Work with Objects in the Toolbox](#)

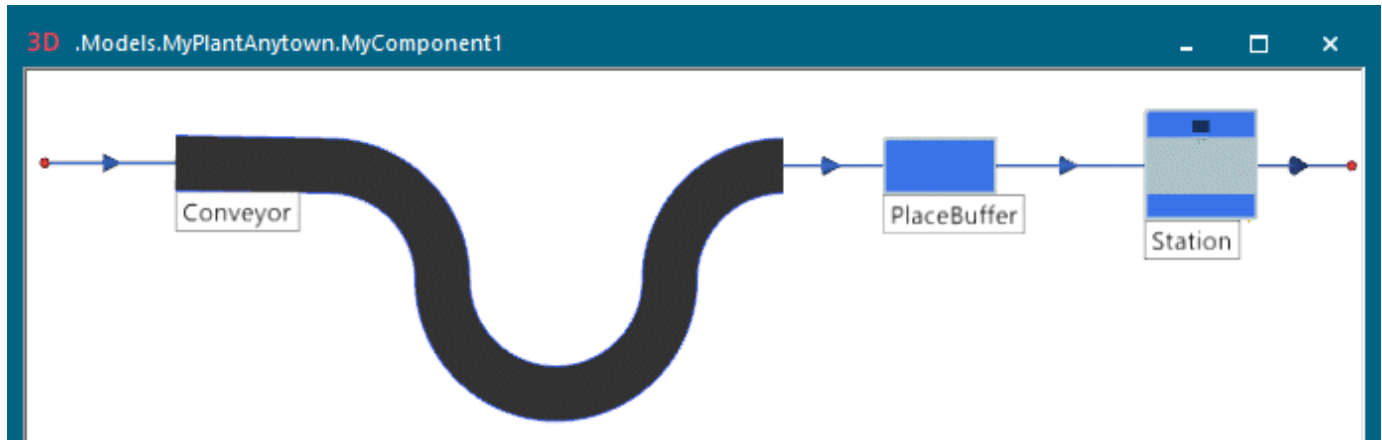
Modeling Hierarchically

Model Hierarchically [general]

Modeling hierarchically is the concept of inserting components, which you modeled in a *Frame*, into other *Frames*. This way you can model, and test, the individual components of your simulation model detached from the *Frame* which holds your complete simulation model.

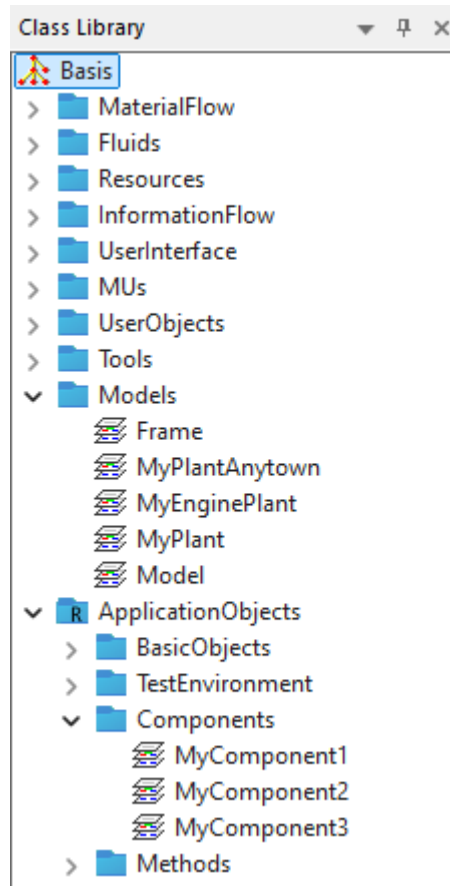


You can combine any number of components in the *Frame* in which you build your overall model. In our example we modeled a component in the *Frame* named **MyComponent1** and inserted it into *Frame MyPlantAnytown*, which holds our complete simulation model.



To open the window of the *Frame* from which the current *Frame* has derived, click . In our example, clicking **Open Origin** opens the *Frame* **MyComponent1** in the folder **ApplicationObjects > Components** in the *Class Library*.

You can thus model the components, such as large machines or entire departments, to match their counterpart in your real-life plant as closely as possible. You can also insert these components several times. You can use these components in exactly the same way as you use any of the other user-defined or built-in objects. You can create icons for them and insert them into your model just as you would any of the material flow objects. In addition it enables you to clearly structure your simulation model.



Another advantage of modeling hierarchically is the fact that you can distribute developing the simulation model among several colleagues, even among several sites. You can then integrate the components into the complete model at any point in time.

Compare [Working with the Frame](#)

Compare [Show the Contents of a Frame in the Class Library](#)

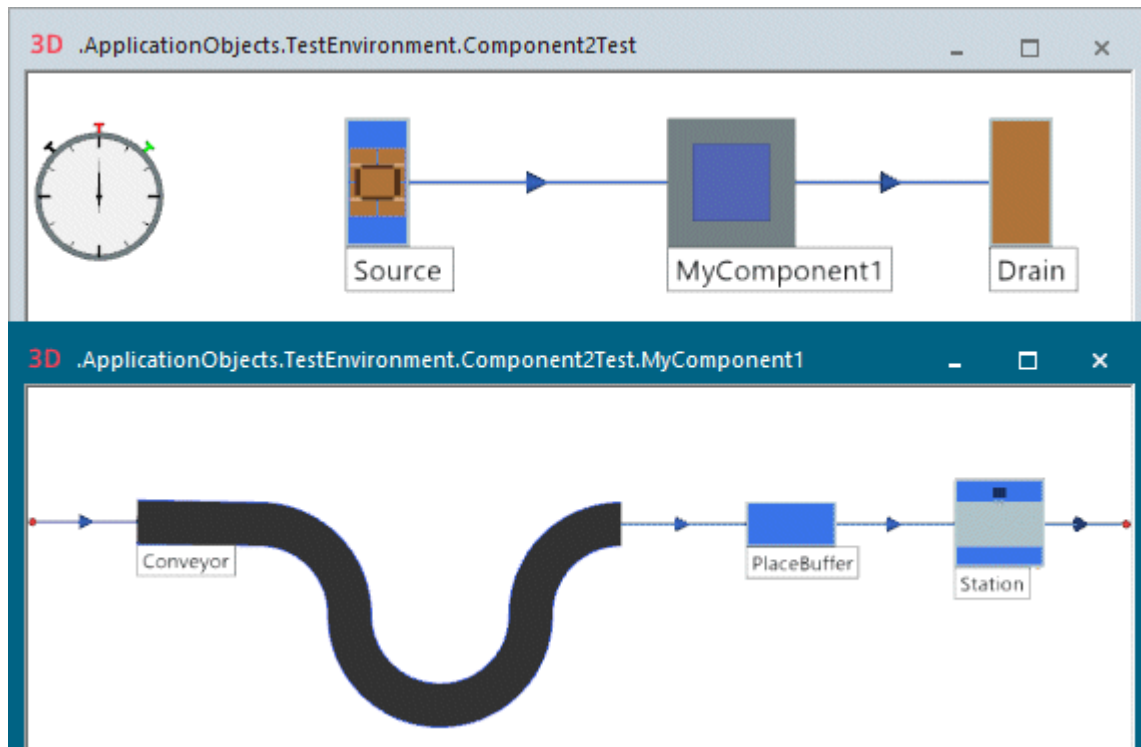
Compare [Model Transitions Between Frames](#)

Compare [Show Reports in a Hierarchically Structured Model](#)

Back to [Creating a Simulation Model](#)

Test a Component You Modeled

Modeling hierarchically enables you to test the components detached from your complete simulation model, thus ensuring that they work as intended, before integrating them. For this you can create a testing environment with the desired level of detail. Insert the component into the test *Frame*, insert a *Source* and a *Drain*, connect all objects and start the simulation.



Back to [Model Hierarchically \[general\]](#)

Working with the Frame

Working with the Frame

As a rule you will create your simulation models in the *Frame*  that we placed in the folder **Models** in the *Class Library*. Naturally, you can also insert additional *Frames* into this folder and model there.

To rename this *Frame*, so that it matches the name of the plant you are modeling, do one of the following:

- Right-click it in the *Class Library* and select **Rename**.
- Select it in the *Class Library* and press **F2**.
- In the *Frame* window itself, click **Rename** on the **Home** ribbon tab or press **F4**.

To build your simulation model, you will insert any of the built-in objects or any objects you yourself design, and connect these objects, which represent your stations, with the *Connector*.

To run the simulation you will insert an *EventController*  that starts, stops and resets the simulation run. You can use each and every simulation model as an application object to create additional simulation models.

By placing *Frames* within *Frames* you can build hierarchically structured models. This way you can build and structure simulation models that exactly match the real world systems you want to simulate. It also

enables you to break down complex tasks into manageable chunks and distribute a laborious task to several co-workers.

Back to [Creating a Simulation Model](#)

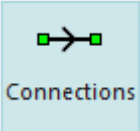
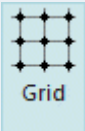
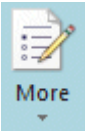
Select View Options in the Frame

You can select general view options as well as options that only apply to the *Frame* in which you are working.

You can select general options for modeling under **File > Preferences/Model Settings** in *Plant Simulation* window. You can also select options for each *Frame* individually, which only apply to that *Frame*.



Click the buttons on the **View** ribbon tab of the *Frame*.

To do this	Click
Hide or show the names of objects in the <i>Frame</i> .	 Names
Hide or show <i>Connectors</i> in the <i>Frame</i> .	 Connections
Hide or show Comment objects in the <i>Frame</i> .	 Comments
Hide or show the grid in the <i>Frame</i> .	 Grid
Set additional view options for the <i>Frame</i> .	 More

Initially any *Frame* you insert into your model uses the settings you selected under **File > Preferences/Model Settings** in the *Plant Simulation* window. Select any of the commands to activate or deactivate

them for the selected *Frame*. To restore the built-in settings, and deactivate any of your changes, select **Inherit Settings**.

Compare [Select Settings in Plant Simulation](#)

Back to [Working with the Frame](#)

Model with Objects from the Class Library

Model with Objects from the Class Library

To create a simulation model, you will insert instances of the class objects from the *Class Library* folders into your model. This, as a rule, will be the *Frame* that is located in the folder **Models** in the *Class Library*.

You can modify and expand the structure of the *Class Library* to meet the requirements of your simulation project. You might, for example, add additional folders to save your simulation models, the test runs and the different components of your model to.

Note

You can rename any object in the *Class Library*. Be aware though that your simulation model may not run correctly any more when you rename an object that you entered as entrance or exit control into another object or *MUs Plant Simulation* entered in a *Source* object under **Attributes > MU Selection > MU**. In these cases the path statements will be wrong after renaming objects.

You can:

- [Insert an Object from the Class Library](#)
- [Insert an Object from the Toolbox](#)

Back to [Working with the Frame](#)

Insert an Object from the Class Library

In the structure, navigate to the folder and then to the object you would like to insert. The objects are grouped according to their functions and to the frequency you use them.

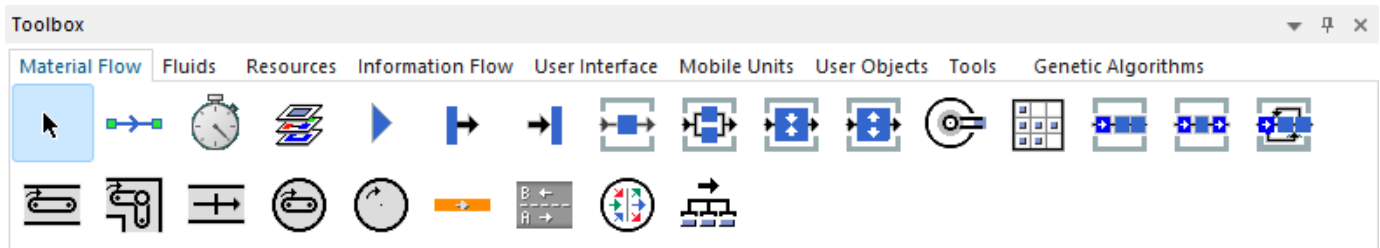
To insert an instance of the class object into the *Frame*:

- Click the object of your choice with the left mouse button to select it. The insert cursor looks like this .
- While holding the mouse button down, drag the object to the target position in the *Frame* window and drop it.

Back to [Model with Objects from the Class Library](#)

Insert an Object from the Toolbox

You can insert objects from the *Toolbox* that contains the objects grouped on tabs/toolbars. You might, for example, insert one of the built-in material flow objects from the toolbar **Material Flow**.



- Click the toolbar, i.e., the tab, which contains the object you would like to insert.
- Move the mouse to the icon of the object.
- Click the object with the left mouse button to select it.
- Move the mouse to the target position in the *Frame* window and click the mouse button once.
- To insert several instances of the same object class, hold down the **Shift** or the **Ctrl** key while you click the mouse button in the *Frame* window. This way you do not have to return to the *Toolbox* each time after inserting an object.
- To go to the class of an object, press the **Ctrl** key and click the object in the *Toolbox*. *Plant Simulation* highlights the class object in the tree window in the *Class Library*. Double-click that object in the tree window to edit class properties or hold down the **Ctrl+Alt** keys and click the object in the *Toolbox*. This does not work, when the object is already selected.

Back to [Model with Objects from the Class Library](#)

Insert Objects into the Frame

You can insert the objects, which symbolize the machines and stations in your plant, into the *Frame* that symbolizes the installation or plant you are modeling.

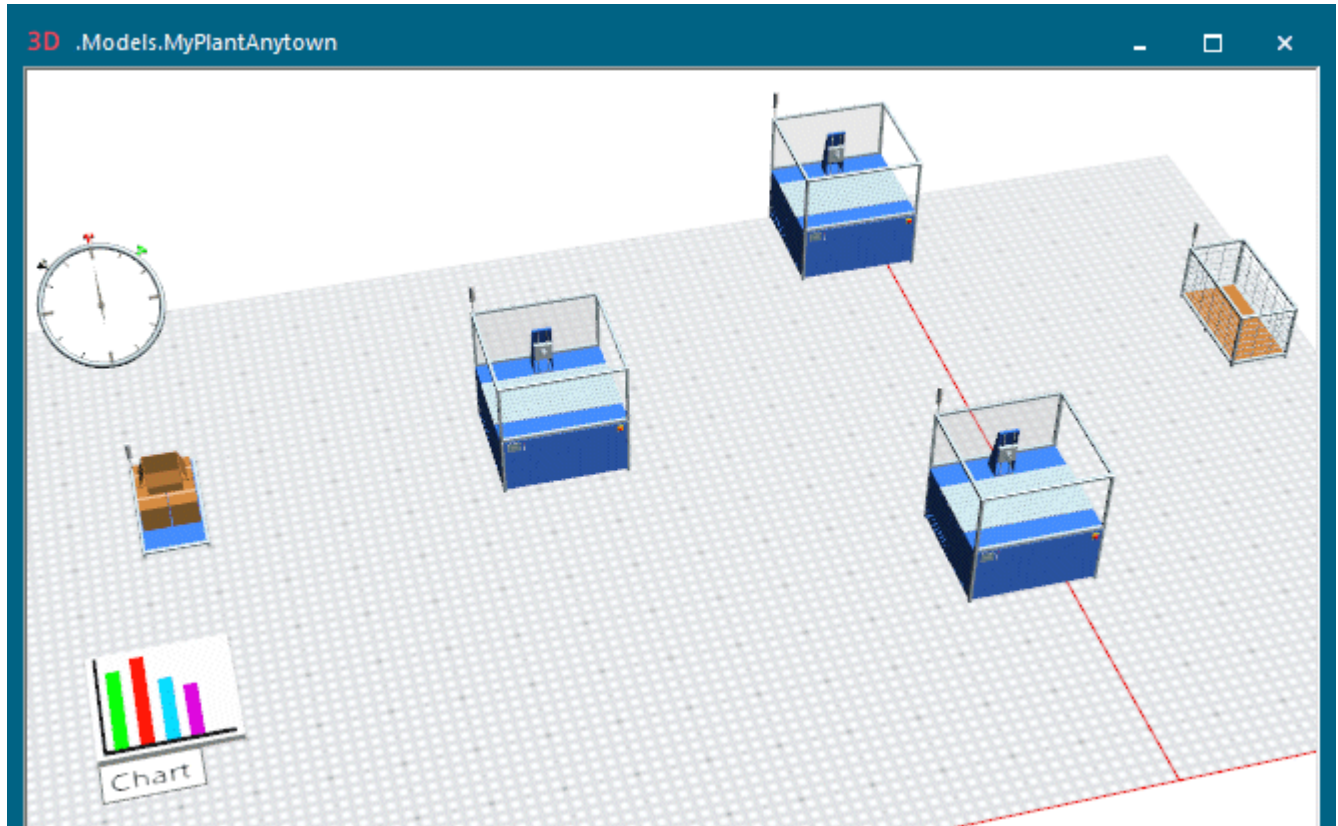
Proceed as follows:

- Start *Plant Simulation* and open a new model: Click **Create New Model** on the **Start Page** or select **File > New**.

This opens the *Class Library* with the built-in objects *Plant Simulation* provides, the *Toolbox* and an empty *Frame*—in the folder **Models**—for creating your simulation model.

- To insert an object into the *Frame*, do one of the following:

- Click the icon of the object you would like to insert in the **toolbox**, move the mouse pointer, which then turns into crosshairs (+), to the location in the *Frame* into which you would like to insert it and click the left mouse button once.
- Use drag-and-drop to insert objects from the **class library** into the *Frame*.
- To insert several instances of an object one after the other, click the object in the *Toolbox*. Hold down **Ctrl** or **Shift**, click the position in the *Frame* at which you would like to insert the first instance and click the left mouse button. Keep holding down **Ctrl** or **Shift** and repeat this until you have inserted the desired number of instances and then release the key.
- Insert the objects, so that your model matches the screenshot.



- Click the folder or the tab **Material Flow** and insert a *Source* , three *Stations* , and a *Drain* .
- Click the folder or the tab **User Interface** and insert a *Chart* .
- To move an object in the *Frame*, do one of the following:
 - Click the object once with the left mouse button, hold the button down, move the icon to the desired location, and release the mouse button.
 - To move the selected object one pixel at a time in the direction of the key, press the respective arrow key.
 - To move the selected object one grid unit at a time, hold down **Shift**, and press the respective arrow key.

- To align all objects to grid points in the *Frame* window, select all objects by dragging a marquee over them and click **Align to Grid** on the **Icons** ribbon tab of the *Frame*.
- To delete an object, press the **Delete** key on the keyboard or click it with the right mouse button and select **Delete**.

Compare [Insert an Object from the Class Library](#)

Compare [Insert an Object from the Toolbox](#)

Compare [Work with Objects in the Frame Window](#)

Back to [Working with the Frame](#)

Connect Objects with the Connector

The objects you insert into your simulation model represent stations in a production process. Each material flow object in your model has entrance points which receive parts, and exit points from where the parts move on to the succeeding objects in the sequence of stations. The *Connector* establishes the material flow connections using these points.

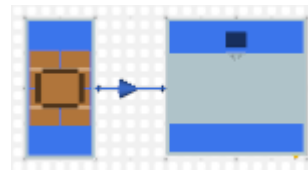
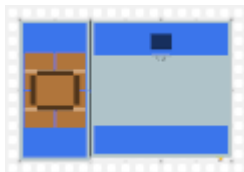
In real installations the relation between the different stations is obvious by the way in which the machines are arranged on the shop floor. *Plant Simulation*, on the other hand, handles this somewhat differently.

To clarify to which other object parts are to be moved when leaving an object, you have to establish connections between these objects.

- **To connect objects automatically**, place them next to each other when inserting them.

This works when [Connect Objects Automatically \[model settings\]](#) is turned on. It works best, when you [Use Planning View](#).

To check if the objects really are connected, select one of them and move it on the grid. Click **Undo** afterward to move it back to the original position.



- **To connect the stations that represent the flow of materials:**

Click the **Connector**  in the *Toolbox* to activate connect mode. The mouse pointer turns into crosshairs (+). When you connect objects in the *Frame*, *Plant Simulation* attaches the connecting line to the front end of the cursor , allowing you to actually see the connection before setting it.

To connect object A and object B, first click object A and then object B.

Note

Plant Simulation only shows the connection, when you select **Show Connections** on the **View** ribbon tab.

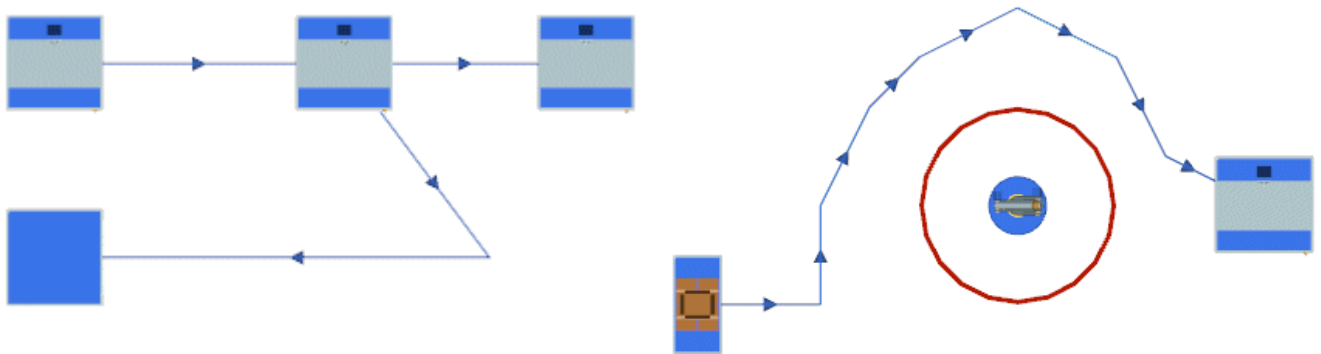
Drag the *Connector* from the *Class Library* or the *Toolbox* onto an object inserted into a *Frame* that you would like to connect. *Plant Simulation* automatically places the starting point of the *Connector* to this object.

- To connect several objects one after the other without having to return to the *Toolbox* each time you established a connection, hold down **Ctrl**, while you establish the connection.
- To automatically connect the objects while you insert them, place the objects right next to each other, when you insert them. You will then have to manually move the objects to their final position.

Note

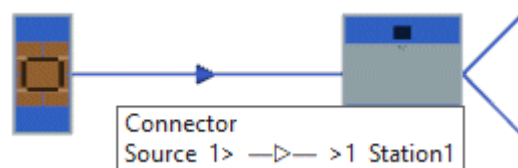
This only works, when **File > Preferences/Model Settings > User Interface > Connect Objects Automatically** is active. In addition the position of the exit of the one and the entrance of the other object are to be no more than 3 pixels apart.

- To tell *Plant Simulation* where and how to place the *Connector* in the *Frame*, hold down **Shift**, **Ctrl**, or **Alt** while you establish the connection:
 - *Connector*, no key: *Plant Simulation* aligns the connection to the grid, i.e., to the grid points next to the location where you click the mouse button.
 - *Connector+Ctrl*: *Plant Simulation* stays in connect mode, so you can connect several objects one after the other without having to return to the *Toolbox* each time you established a connection.
 - *Connector+Shift*: *Plant Simulation* inserts the connection at a right angle, regardless of where you click.
 - *Connector+Alt*: *Plant Simulation* sets anchor points of the connection at the location where you click. This is similar to the freehand drawing feature.
- To create a **non-straight connection**, set one or more anchor points: First, click object A, proceed to point 1 and click once, then proceed to point 2, click once, and finally move to object B and click the mouse button once to establish the connection.

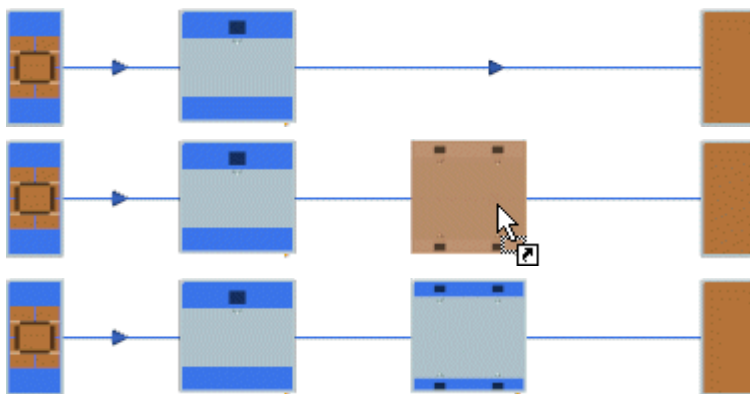


- To create a **right-angled connection**, hold down **Shift** and click the left mouse button to set the anchor point.

- To move an anchor point, click it, hold the mouse button down, and drag the handle  to the desired location. *Plant Simulation* also moves the *Connector*.
- To check if all connections between the objects really have been established, click **Unconnected Objects** on the **View** ribbon tab. *Plant Simulation* selects all objects with unconnected entrance and exit points.
- To make the connecting line thicker, enter a higher number into the text box **Weight**.
To make the connecting line thinner, enter a lower number into the text box **Weight**.
- To select another color than the default black, in the dialog **Colors**, click the field next to **Color**.
You can select one of the predefined colors, or you can click **More Colors** and click the **Select** button to select a color in the color matrix. Then click **OK**. *Plant Simulation* show this color next to **More Colors** and uses it as the active color.
- To show the source and the target of the *Connector* in the *Frame* is a *Tooltip*, drag the mouse over the *Connector*.

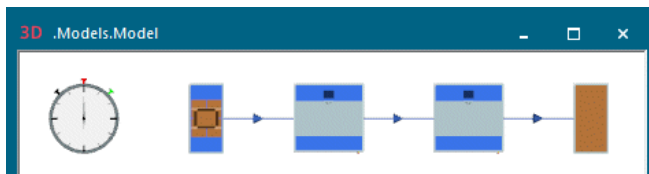
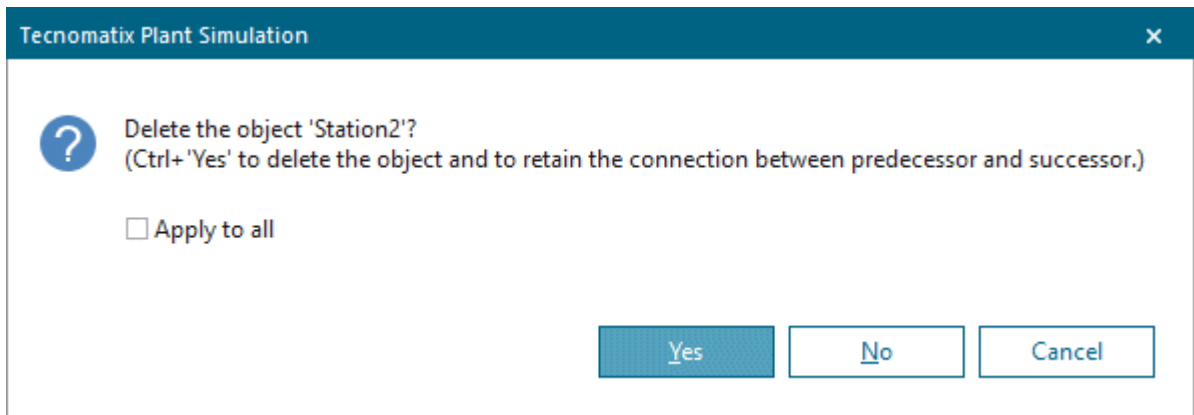


- To **exchange the successor** of a *Connector*, select the end point of the *Connector* and drag it to another object.
To **exchange the predecessor** of a *Connector*, select the starting point of the *Connector* and drag it to another object
- To **place an object**, which you insert from the *Class Library* into a *Frame*, **in between two already connected objects**, drag that object onto that spot of the *Connector* and drop it there. In this process *Plant Simulation* exchanges the successor of the original *Connector*.

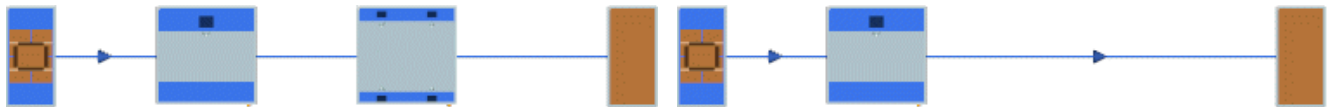


This also works when you drag a *Frame*, which contains exactly one suitable **entrance Interface** and exactly one suitable **exit Interface**, onto the *Connector*.

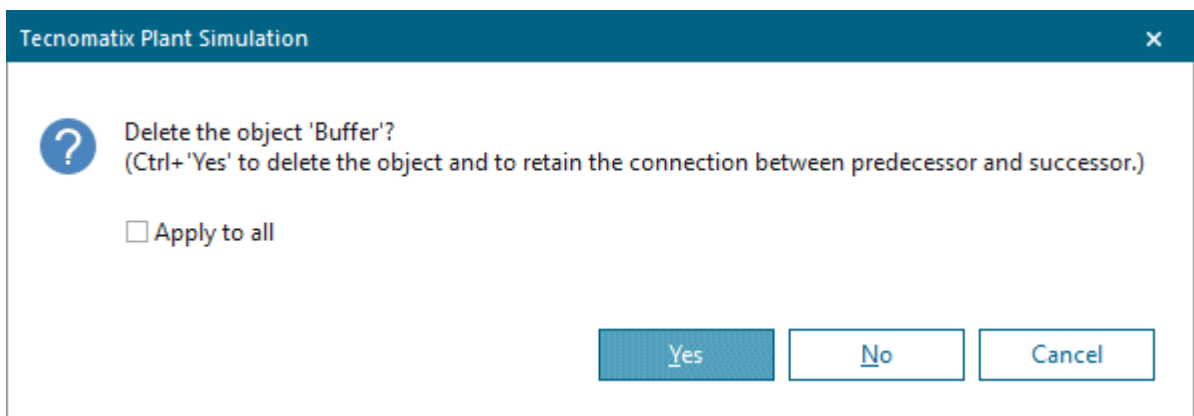
- To retain the *Connector* between objects when you delete an object, hold down **Ctrl** and then press **Del** while keeping **Ctrl** pressed. Click **Yes** in the dialog that opens.



- To **automatically connect predecessor and successor when deleting an object** located between other connected objects, hold down **Ctrl** while you delete the object with the context menu command **Delete**.



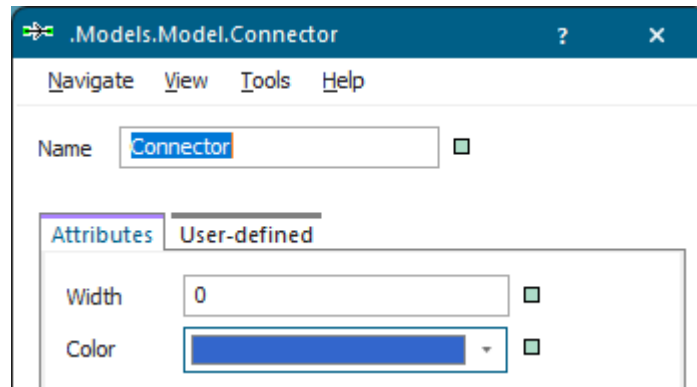
When you work with the **Delete** key, select the object(s) you want to delete, and press **Delete**. Follow the instructions in the dialog that opens.



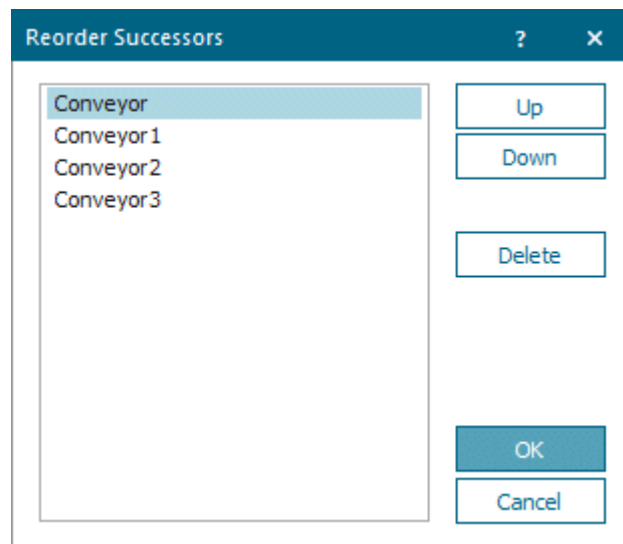
Note

This does not work when an object has more than one predecessor and successor. If the object has a successor, *Plant Simulation* exchanges the successor of all preceding *Connectors*. If the object has a predecessor, *Plant Simulation* exchanges the predecessor of all succeeding *Connectors*.

- To change the **Width** and the **Color** of the *Connector*, double-click its icon and enter and select the settings you need.



- To **reorder the sequence of the succeeding objects**, i.e., the sequence of the *Connectors*, click the object with the right mouse button and select **Reorder Successors**. Move the selected successor with the **Up** or **Down** buttons.



Compare [Auto-connect Length-oriented Objects](#)

Compare [Undo \[3D\]](#) and [Redo \[3D\]](#)

Back to [Working with the Frame](#)

Add a Background Graphic to the Frame

You can add a background graphic to the background of the *Frame*.

To add a graphic to the background of a *Frame*:

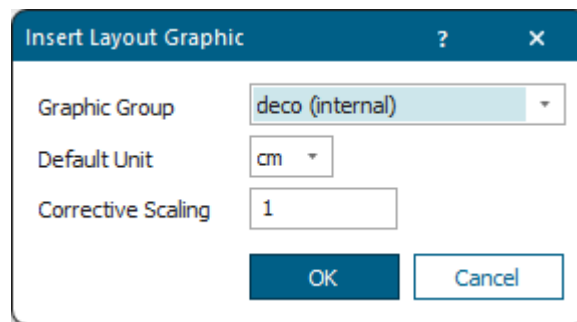
- Drag a .gif, .bmp, .ppm, .ppm raw, .dgn, .dxf, or a .dwg file from the Windows Explorer or the Internet Browser over the background of the *Frame* and drop it there.
- Drag a layout drawing over the background of the *Frame* and drop it there.

You can use this layout drawing as the actual background of the *Frame* and insert the *Plant Simulation* objects on top of it. This is feasible when your simulation model is not too complex.

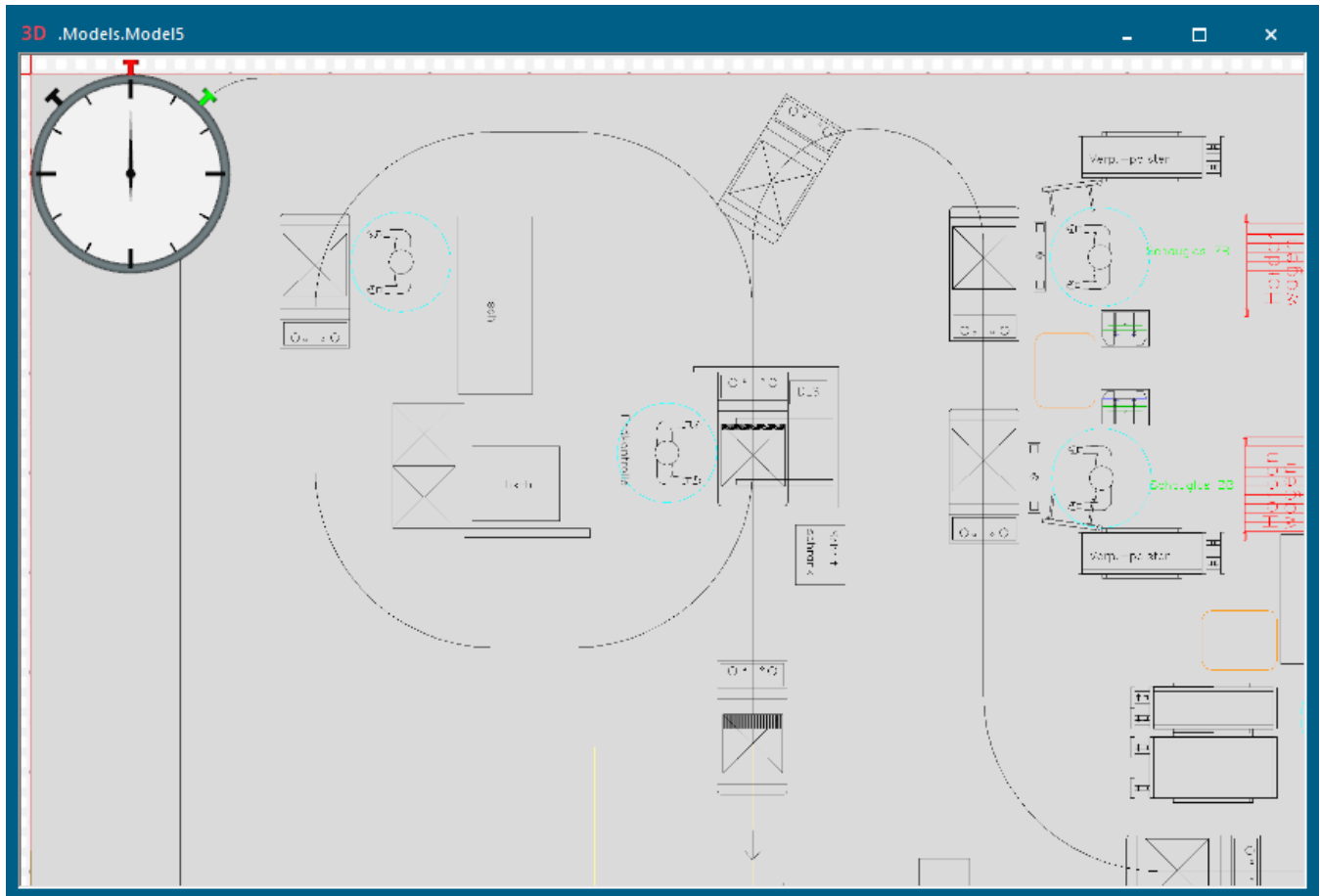
Note

If you are modeling a complex plant, this might not be practicable, as it might lead to a very cluttered layout.

Be aware that the .dxf, .dgn or .dwg files are vector-based, whereas *Plant Simulation* uses pixel-based graphics. This means that you have to think about how to transfer one size to the other.



If the CAD-drawing contains units and its dimension is not greater than 32000 by 32000 pixels, *Plant Simulation* automatically uses a **Corrective Scaling** of 1.



If the CAD-drawing contains units and its dimension is greater than 32000 by 32000 pixels, *Plant Simulation* scales the drawing so that it fits into the *Frame* window.

The *Console* window shows the origin of the dwg drawing and the dimension of the drawing.

Console

```
D:\Documentation\MyData\FilesForHelp\CAD Dateien für 3D\Plan-Bd20_x0_y0.dwg:
Drawing origin: 14.3052052905911, 15.1654516195426 [CAD units]
Drawing dimension: 28.6104105811821, 30.3309032390852 [CAD units]
```

Compare [Add a JT Layout File to Your Simulation Model](#)

Compare [Import a 3D Graphic](#)

Compare [Add a Point Cloud](#)

Back to [Working with the Frame](#)

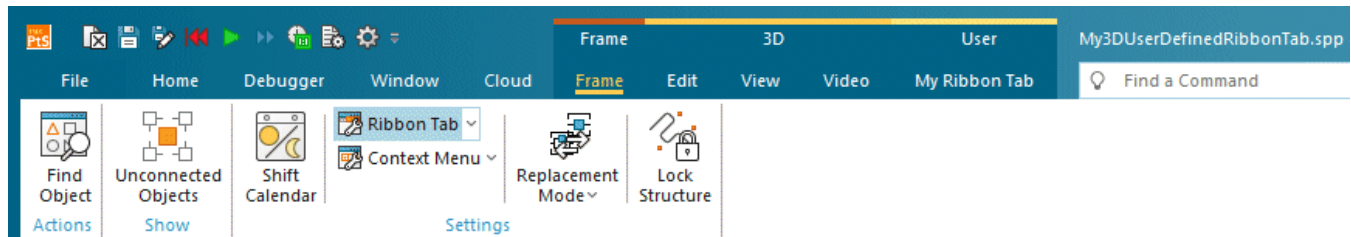
Create Your Own Ribbon Tab/Context Menu in the Frame

You can create your own ribbon tab/context menu in the *Frame*.

To create a *user-defined ribbon tab* or a *user-defined context menu* with your own, frequently used, ribbon commands or context menu commands respectively:

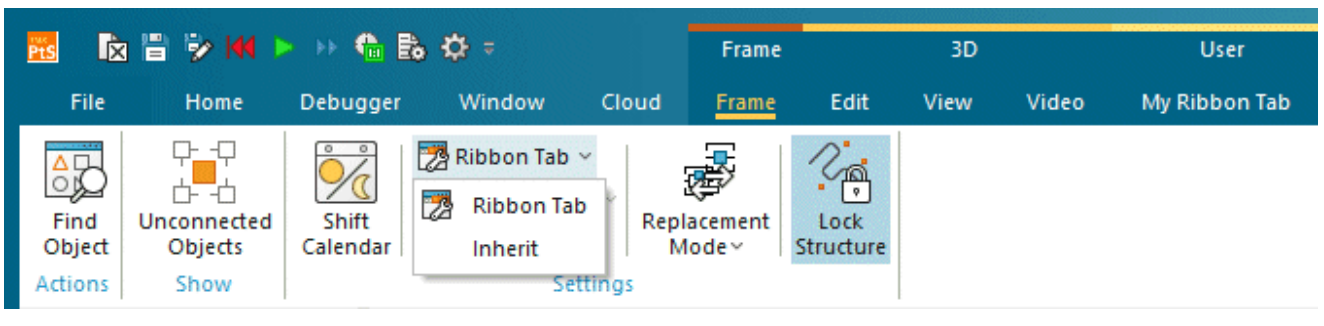
- Click **Configure User-defined Ribbon Tab** or **Configure User-defined Context Menu** on the **Frame** ribbon tab of the *Frame* for which you want to create the *user-defined ribbon tab* or *user-defined context menu*.

If a *user-defined ribbon tab* exists, the button on the **Frame** ribbon tab is highlighted.

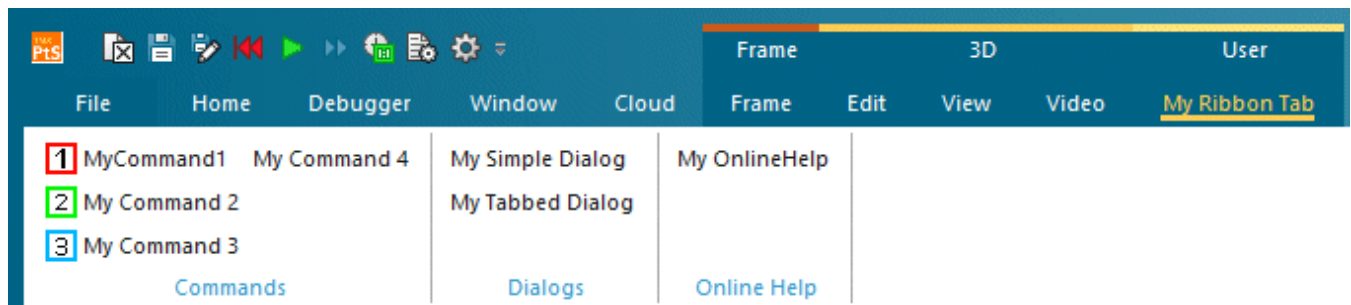


Note

To create a *user-defined ribbon tab* in a *Frame*, which you inserted into another *Frame*, deactivate the command **Inherit**, so that it does not show a check mark to the left.



- Type in the **Title** of the *user-defined ribbon tab* with which *Plant Simulation* shows the *user-defined ribbon tab*. We typed in **My Ribbon Tab**.



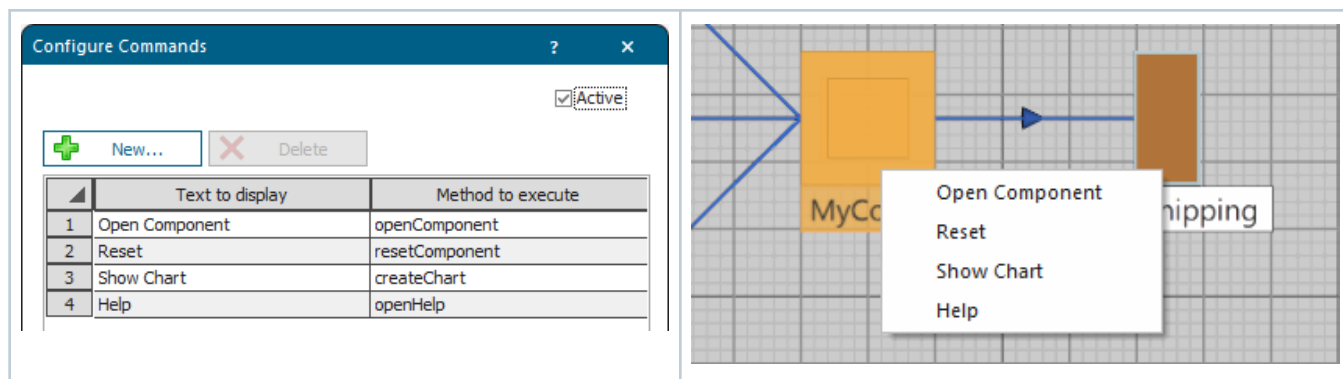
- To show the *user-defined ribbon tab* for the *Frame*, for which you defined it, select **Active**.
- Type in the name of the **command** that *Plant Simulation* shows on the *user-defined ribbon tab* in the *Frame*. Type an & (ampersand) in front of a letter to make this letter the access key. You can select this

ribbon command by holding down **Alt**, by pressing the predefined key combination **U** and **T**, and by the pressing that letter. The built-in *Plant Simulation* access keys take precedence over any access keys you define!

You can also type in a formula as a ribbon command or a context menu command. A formula is designated by a leading question mark. When you type in `?Method1` for example, the method named **Method1** will be called. The return value of this method has to be of data type *string*. This way you can switch between different texts, for example between **Activate** and **Deactivate** and you can translate the ribbon or context menu commands into different languages. If the methods return an empty string "", *Plant Simulation* hides the respective ribbon or context menu command.

Note

You can use any formulas, even for example a method call with parameters, such as `?Method1(42)` or a table access, such as `?DataTable[1, 3]`.



- Type in the *Methods*, which the ribbon/context menu commands execute into the text boxes below **Method to execute**.

If the method expects a parameter, *Plant Simulation* automatically passes it. To this parameter, which has to be of data type *list*, the selected objects, which are located within the same *Frame* is the *Frame* whose ribbon tab or context menu is opened, are passed.

When *Methods* are called from the *user-defined ribbon tab*, the anonymous identifier `?` (question mark) points to the *Frame* in which you selected the command. This way you can access objects within the respective *Frame*.

Note

The commands on the *user-defined context menu* only apply to the selected object(s) within the *Frame* for which you defined the context menu.

The *Class Library* does not show the *user-defined context menu*.

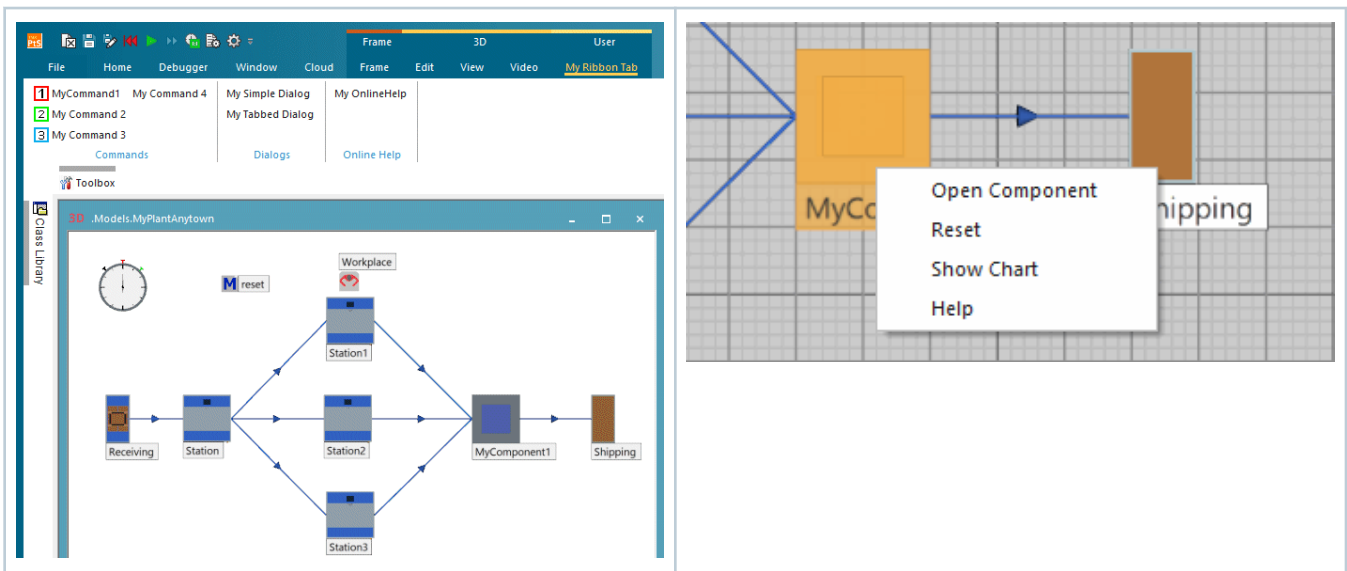
User-defined ribbon tab of a Frame

	Text to display	Method to execute
1	#Commands	self.OnClicked
2	MyCommand1	myAction1
3	My Command 2	myAction2
4	My Command 3	myAction3
5	My Command 4	myAction4
6	#Dialogs	
7	My &Simple Dialog	openMySimpleDialog
8	My &Tabbed Dialog	openMyTabbedDialog
9	#Online Help	
10	My OnlineHelp	openMyHelp

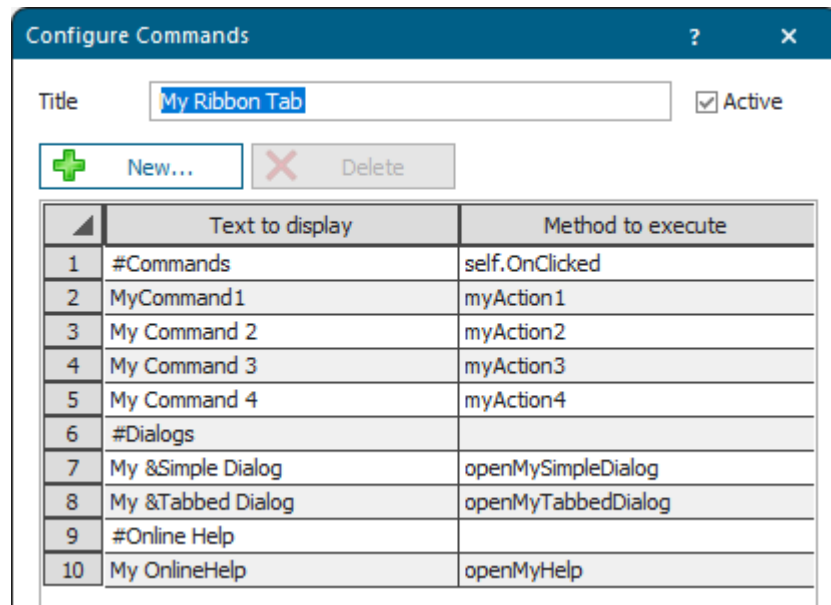
User-defined context menu of a Frame

	Text to display	Method to execute
1	Open Component	openComponent
2	Reset	resetComponent
3	Show Chart	createChart
4	Help	openHelp

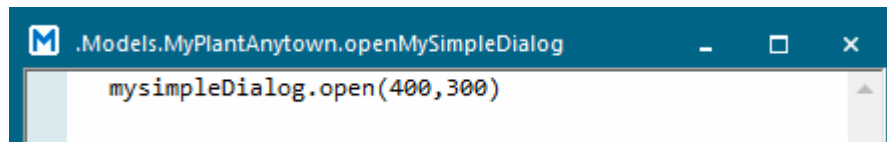
- To create the *user-defined ribbon tab* or the *user-defined context menu*, click **OK**. *Plant Simulation* shows the ribbon tab to the right of the pre-defined ribbon tabs.



- To open a user-defined dialog from the *user-defined ribbon tab*:
 - Type in the name of the command, which opens the user-defined dialog, into the text box below **Text to display**.
 - Click  to enter the path to and the name of the *Method M*, which tells *Plant Simulation* to open the dialog, into the text box below **Method to execute**.



- Create a *Method* with the name you typed in above and type in `name_of_your_user_defined_dialog.open`.

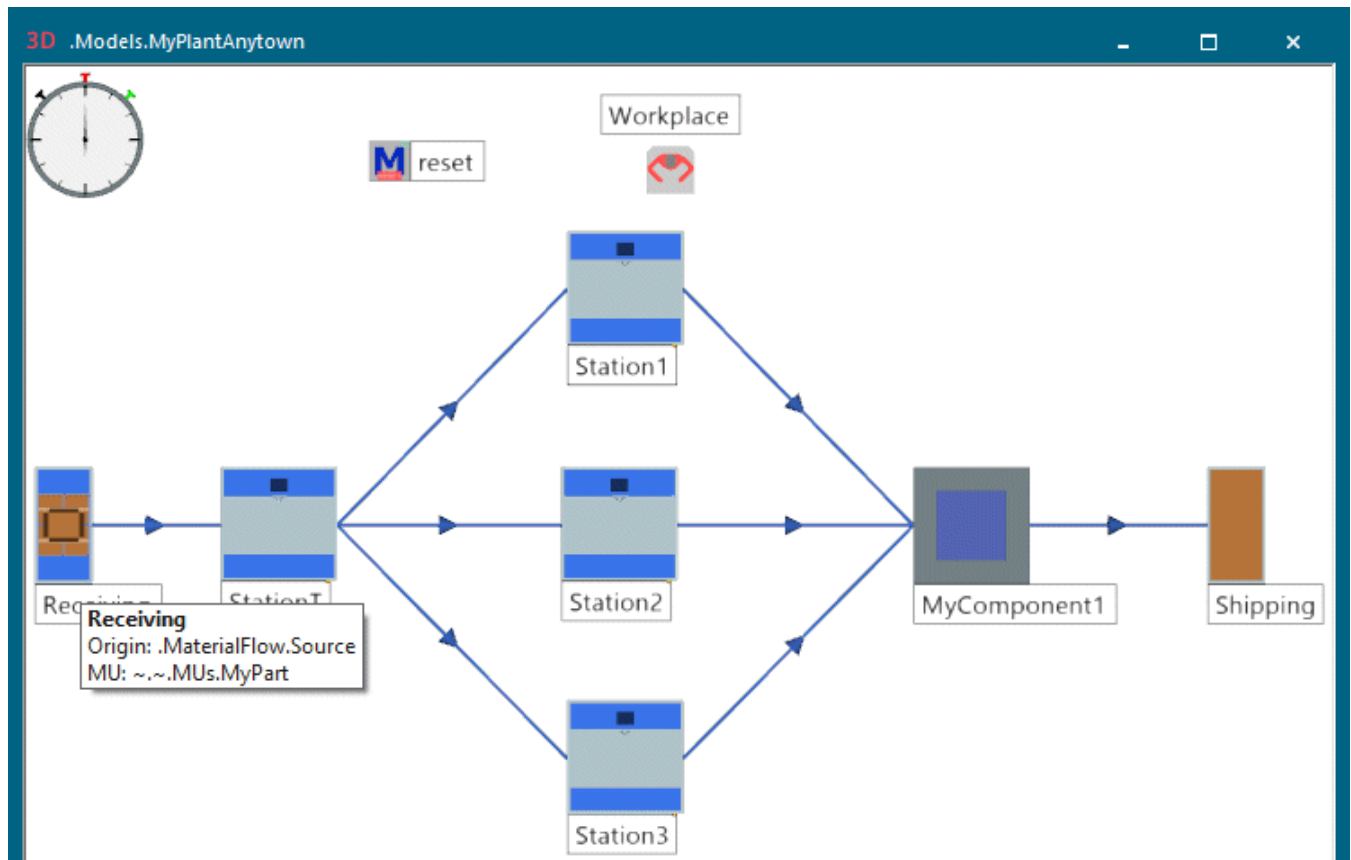


Back to [Working with the Frame](#)

Work with Objects in the Frame Window

You can manipulate the objects, **simulation objects** as well as vector graphics you drew and inserted into the *Frame*.

- To open the object or the *Frame* window, double-click the object or a *Frame* in the model *Frame*, *MyPlantAnytown* in our sample model.
- To insert an object in the *Frame*, select the object in the *Class Library* or the *Toolbox*, drag it over the *Frame* window and drop it there.
- To show or hide the grid in the *Frame* window, click  **Grid**. Showing the grid makes it easier to insert an object at exactly the position you need it to be at.
- To select an object, click it with the left mouse button. You can then move it in the *Frame* by dragging the mouse.
- To nudge the selected object one pixel at a time in a direction, press the respective arrow keys on the keyboard. To move the selected object one grid unit at a time, press **Shift**+ the respective **arrow** key.
- To deselect a selected object, click another object or click anywhere in the *Frame*.



- To select more than one object in a *Frame* at a time, hold down **Shift**, and click the objects you want to mark or drag a marquee over the objects: Hold the left mouse button down, and drag the mouse over the objects, making sure the marquee encloses all of them, and then release the mouse button. You can also combine both methods.

To select all objects in the *Frame* window, press **Ctrl+A**.

- To search for any object in the *Frame* window, click **Incremental Find Object** on the **Frame** ribbon tab and start typing its name. *Plant Simulation* then finds and selects the object.
- To connect objects, use the *Connector*. The *Connector* displays handles  after you select it. Click a handle to change its shape, i.e., to make the *Connector* longer or shorter or to move the anchor point up or down.
- To show or hide connections between objects which you established with the *Connector*, click  on the **View** ribbon tab.
- To show objects that are not connected to another object, click  on the **Frame** ribbon tab.
- To zoom the contents of the *Frame* in, roll the mouse wheel forward.
- To zoom the contents of the *Frame* out, roll the mouse wheel backward.
- To show or hide the names of the objects you inserted into the *Frame*, click .

- To prevent another user from modifying the structure of the model, click .
- To edit the icons of the *Frame*, and to add animation structures to it, click  on the **Home** ribbon tab.
- To model transitions between *Frames* use the *Interface*. When you connect *Frames*, which have several *Interfaces*, with objects, *Plant Simulation* opens the dialog **Select Interface**. Select the *Interface* the *Connector* attaches to and click **OK**.

Note

If a *sub-Frame* only contains a single object, you can also connect the *sub-Frame* without having to use *Interfaces*.

- To open the *Frame* you double-click and to close the *Frame* it is contained in, press **Shift** and double-click a *Frame* located within a *Frame*.
- To move down in the hierarchy of *Frames*, closing the active *Frame*, and opening the next *Frame* one level down in the structure, select **Down one Level** on the context menu.
- To open the window of the *Frame* from which the current *Frame* has derived, click **Open Origin**  on the **Home** ribbon tab.
- To close the active *Frame* and open its location, click **Open Location** .
- To assign the different states a *Frame* can be in, use the attributes **StateBlocked**, **StateEntryShut**, **StateResourceMissing**, **StateSetup**, **StateWorking**, **Stopped**, and **Unplanned**. *Plant Simulation* shows the **state** of the object as a colored ring on a pole by default.

Color	Means that the Frame is
red	Failed
pink	Stopped
blue	Paused
light blue	Unplanned
green	Working
yellow	Blocked
brown	Setting-Up
cyan	recovering or has a closed entrance
orange	Waiting for a resource (<i>Exporter</i>)

- To open help for the selected object, press **F1**.

Back to [Working with the Frame](#)

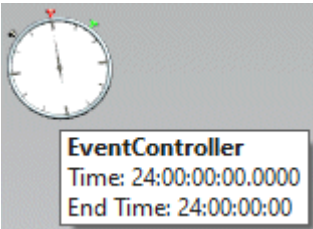
Show Tooltips of Objects in the Frame

To show a *tooltip* with information about an object, drag the mouse over it and hover.

All objects show the **name** of the object in bold face and its **origin**. Depending on the state and the type of the object they show additional information.

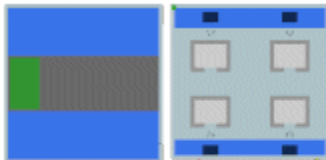
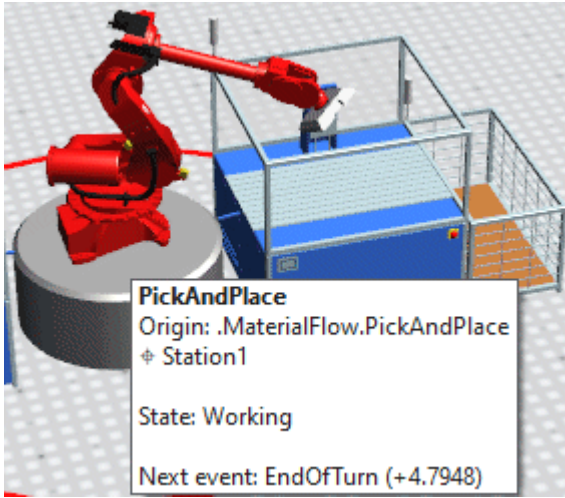
- **Tooltip of the EventController**

The *EventController* shows the **current simulation time** and the **End Time** that you entered.

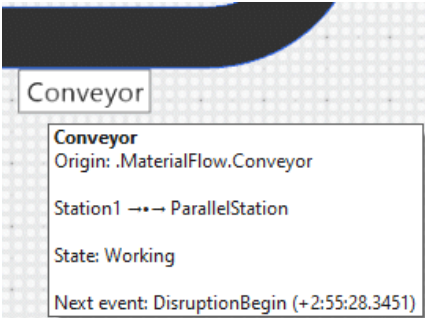
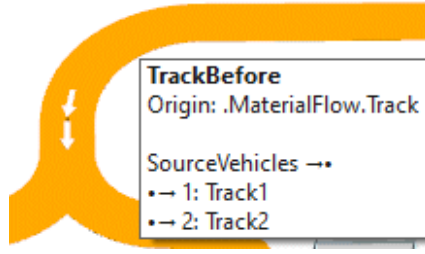
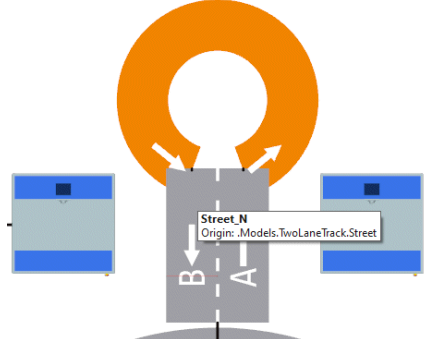


- **Tooltip of Point-oriented Objects**

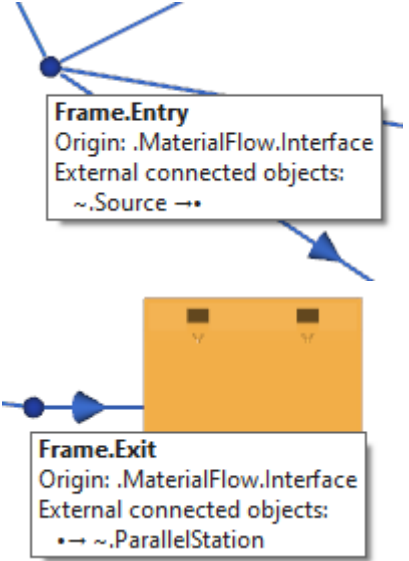
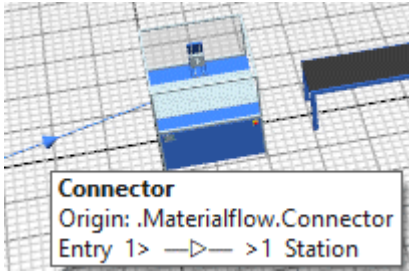
Source	Station	Station Showing Next event
A screenshot of a tooltip for a Source object. It shows a blue and brown rectangular icon. The text in the tooltip is: "Source", "Origin: .MaterialFlow.Source", "MU: *.MUs.Part", and "State: Blocked".	A screenshot of a tooltip for a Station object. It shows a blue and brown rectangular icon. The text in the tooltip is: "Station", "Origin: .MaterialFlow.Station", and "State: Working".	A screenshot of a tooltip for a Station object showing the next event. It shows a blue and brown rectangular icon next to a black conveyor belt icon. The text in the tooltip is: "Station1", "Origin: .MaterialFlow.Station", "State: Working", and "Next event: DisruptionBegin (+ 1:36:55.7240)".

Buffer Showing Label	PickAndPlace Showing Deposit Station
 Buffer1 [Fill level Fill level] Origin: .MaterialFlow.Buffer Fill level Fill level	 PickAndPlace Origin: .MaterialFlow.PickAndPlace + Station1 State: Working Next event: EndOfTurn (+4.7948)

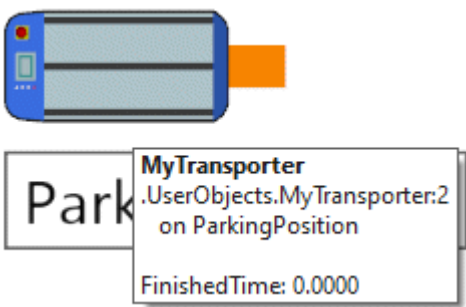
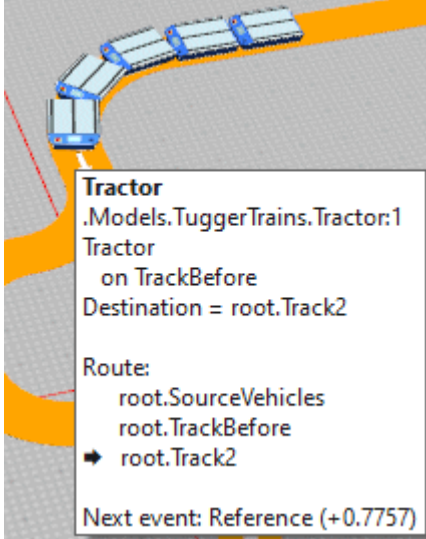
- Tooltip of Length-oriented Objects

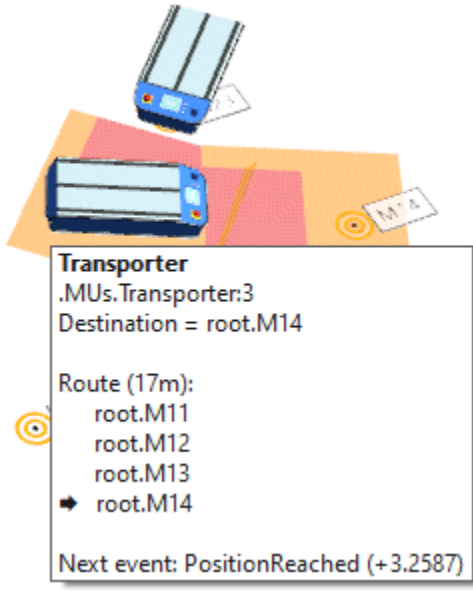
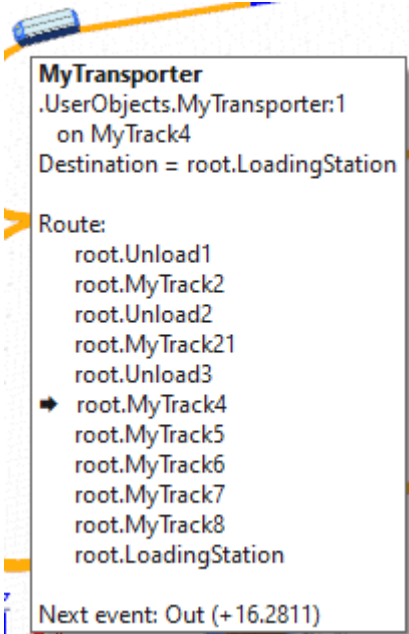
Conveyor Showing Predecessor and Successor	Track Showing Connected Tracks	TwoLaneTrack Showing Sensors
 Conveyor Conveyor Origin: .MaterialFlow.Conveyor Station1 →→ ParallelStation State: Working Next event: DisruptionBegin (+2:55:28.3451)	 TrackBefore Origin: .MaterialFlow.Track SourceVehicles →→ •→ 1: Track1 •→ 2: Track2	 Street_N Origin: .Models.TwoLaneTrack.Street B A

- Tooltip of Interface and Connector

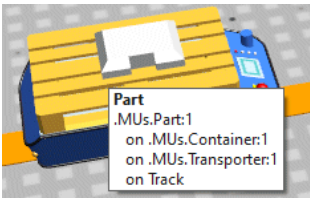
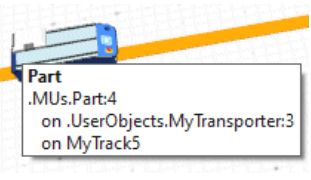
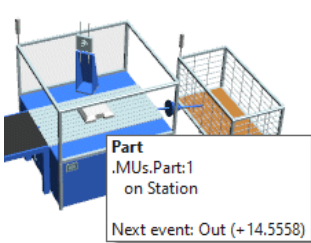
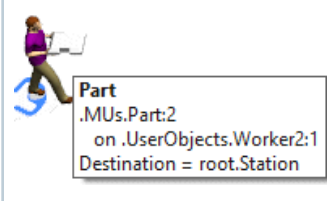
Interface Showing Connected Objects	Connector Showing Connections
 Frame.Entry Origin: .MaterialFlow.Interface External connected objects: ~.Source →• Frame.Exit Origin: .MaterialFlow.Interface External connected objects: •→ ~.ParallelStation	 Connector Origin: .Materialflow.Connector Entry 1> →> >1 Station

- Tooltip of the Transporter

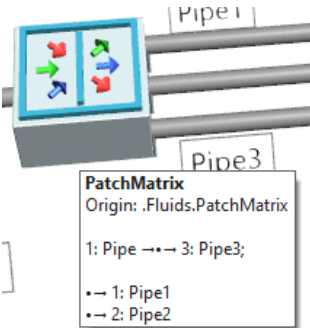
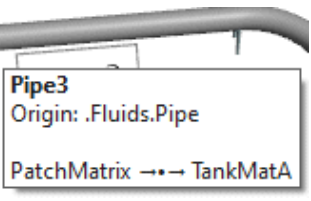
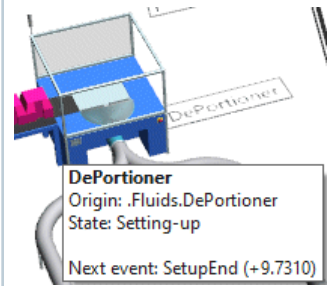
Transporter on Track	Tractor of Tugger Train with Route
 MyTransporter .UserObjects.MyTransporter:2 on ParkingPosition FinishedTime: 0.0000	 Tractor .Models.TuggerTrains.Tractor:1 Tractor on TrackBefore Destination = root.Track2 Route: root.SourceVehicles root.TrackBefore ➔ root.Track2 Next event: Reference (+0.7757)

Transporter Showing Length of Route	Transporter Showing Route
 Transporter .MUs.Transporter:3 Destination = root.M14 Route (17m): root.M11 root.M12 root.M13 ➔ root.M14 Next event: PositionReached (+3.2587)	 MyTransporter .UserObjects.MyTransporter:1 on MyTrack4 Destination = root.LoadingStation Route: root.Unload1 root.MyTrack2 root.Unload2 root.MyTrack21 root.Unload3 ➔ root.MyTrack4 root.MyTrack5 root.MyTrack6 root.MyTrack7 root.MyTrack8 root.LoadingStation Next event: Out (+16.2811)

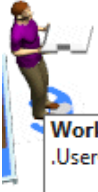
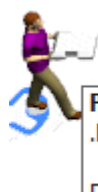
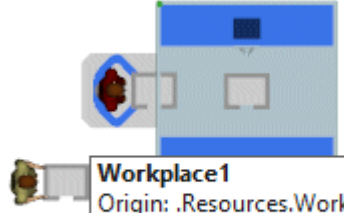
- Tooltip Showing the Stacking Order of Mobile Objects

Part on Pallet on Transporter	Part on Transporter	Part on Station	Part Carried by Worker
 Part .MUs.Part:1 on .MUs.Container:1 on .MUs.Transporter:1 on Track	 Part .MUs.Part:4 on .UserObjects.MyTransporter:3 on MyTrack5	 Part .MUs.Part:1 on Station Next event: Out (+14.5558)	 Part .MUs.Part:2 on .UserObjects.Worker2:1 Destination = root.Station

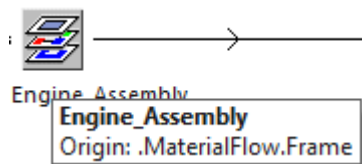
- Tooltip of the Fluid Objects

PatchMatrix Showing Connected Pipes	Pipe Showing Connected Objects	Tank Showing Next Event	DePortioner Showing State
 PatchMatrix Origin: .Fluids.PatchMatrix 1: Pipe →→ 3: Pipe3; •→ 1: Pipe1 •→ 2: Pipe2	 Pipe3 Origin: .Fluids.Pipe PatchMatrix →→ TankMatA	 TankMatB Origin: .Fluids.Tank Next event: TankFull (+20.41.8611)	 DePortioner Origin: .Fluids.DePortioner State: Setting-up Next event: SetupEnd (+9.7310)

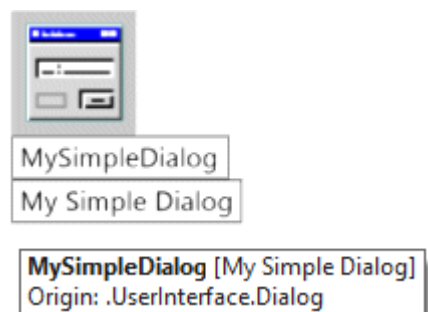
- Tooltip of the Worker and the Worker on Workplace

Part Carried by Worker with Route	Part Carried by Worker Destination	Workplace Assigned to Station Workplace Reserved for Worker (filled arrow) Workplace Reserved for Worker on his Way (hollow arrow)
 Worker2 .UserObjects.Worker2:1 Route (3.7m): root.Workplace root.Workplace1 Next event: PositionReached (+2.4505)	 Part .MUs.Part:2 on .UserObjects.Worker2:1 Destination = root.Station	 Workplace1 Origin: .Resources.Workplace ⇒ Station @ Entrance ◆ .UserObjects.Worker3:1

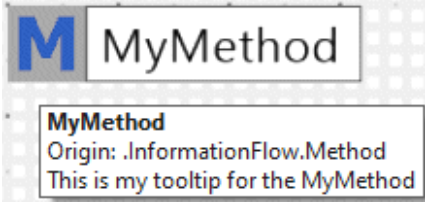
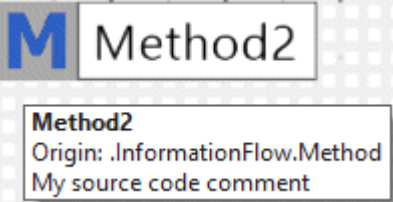
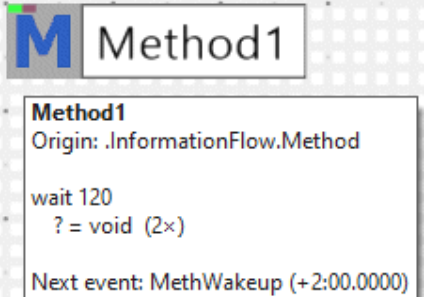
- Tooltip of a SubFrame



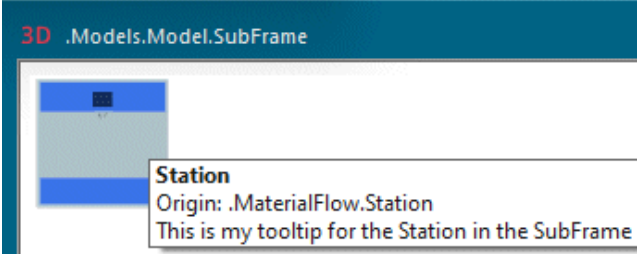
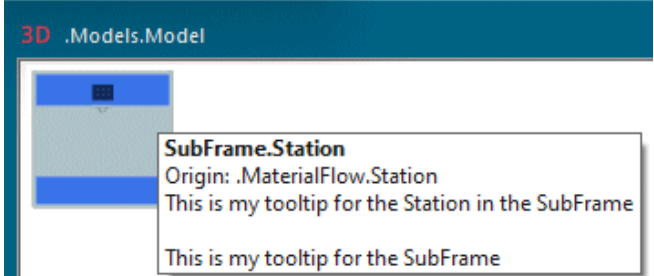
- Tooltip of the object Dialog



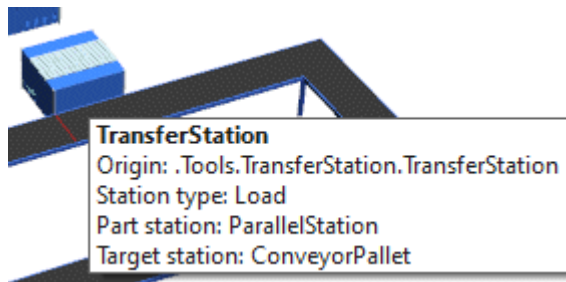
- User-defined Tooltip of a Method

Method with Tooltip as User-defined Attribute	Method with Tooltip of Source Code Comment	Suspended Method
 MyMethod Origin: .InformationFlow.Method This is my tooltip for the MyMethod	 Method2 Origin: .InformationFlow.Method My source code comment	 Method1 Origin: .InformationFlow.Method wait 120 ? = void (2×) Next event: MethWakeup (+2:00.0000)

- Tooltip of a User-defined Attribute in a Frame

User-defined Attribute as Tooltip in SubFrame	User-defined Attribute as Tooltip in Model Frame
 Station Origin: .MaterialFlow.Station This is my tooltip for the Station in the SubFrame	 SubFrame.Station Origin: .MaterialFlow.Station This is my tooltip for the Station in the SubFrame This is my tooltip for the SubFrame

- Tooltip of the TransferStation



Compare [View Information About the Objects](#)

Compare [User-defined Attributes \[general description\]](#)

Compare [Create a Tooltip as a User-defined Attribute](#)

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Connect Objects with the Connector

The objects you insert into your simulation model represent stations in a production process. Each material flow object in your model has entrance points which receive parts, and exit points from where the parts move on to the succeeding objects in the sequence of stations. The *Connector* establishes the material flow connections using these points.

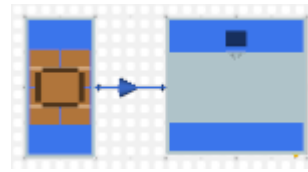
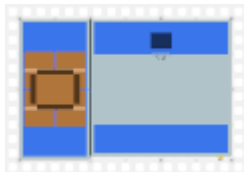
In real installations the relation between the different stations is obvious by the way in which the machines are arranged on the shop floor. *Plant Simulation*, on the other hand, handles this somewhat differently.

To clarify to which other object parts are to be moved when leaving an object, you have to establish connections between these objects.

- **To connect objects automatically**, place them next to each other when inserting them.

This works when **Connect Objects Automatically [model settings]** is turned on. It works best, when you **Use Planning View**.

To check if the objects really are connected, select one of them and move it on the grid. Click **Undo** afterward to move it back to the original position.



- **To connect the stations that represent the flow of materials:**

Click the **Connector**  in the *Toolbox* to activate connect mode. The mouse pointer turns into crosshairs (+). When you connect objects in the *Frame, Plant Simulation* attaches the connecting line to the front end of the cursor , allowing you to actually see the connection before setting it.

To connect object A and object B, first click object A and then object B.

Note

Plant Simulation only shows the connection, when you select **Show Connections** on the **View** ribbon tab.

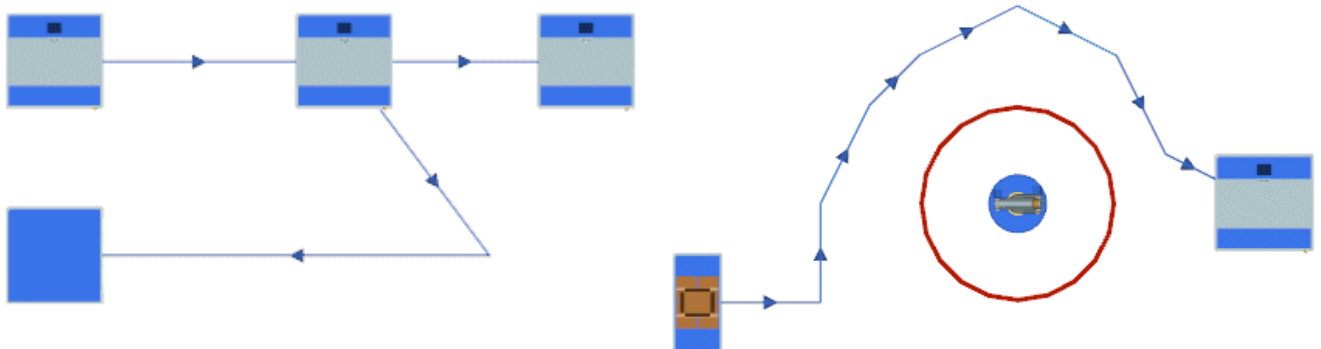
Drag the *Connector* from the *Class Library* or the *Toolbox* onto an object inserted into a *Frame* that you would like to connect. *Plant Simulation* the automatically places the starting point of the *Connector* to this object.

- **To connect several objects one after the other** without having to return to the *Toolbox* each time you established a connection, **hold down Ctrl**, while you establish the connection.
- To automatically connect the objects while you insert them, place the objects right next to each other, when you insert them. You will then have to manually move the objects to their final position.

Note

This only works, when **File > Preferences/Model Settings > User Interface > Connect Objects Automatically** is active. In addition the position of the exit of the one and the entrance of the other object are to be no more than 3 pixels apart.

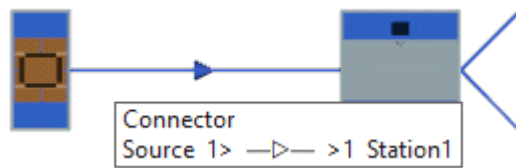
- To tell *Plant Simulation* where and how to place the *Connector* in the *Frame*, hold down **Shift**, **Ctrl**, or **Alt** while you establish the connection:
 - **Connector**, no key: *Plant Simulation* aligns the connection to the grid, i.e., to the grid points next to the location where you click the mouse button.
 - **Connector+Ctrl**: *Plant Simulation* stays in connect mode, so you can connect several objects one after the other without having to return to the *Toolbox* each time you established a connection.
 - **Connector+Shift**: *Plant Simulation* inserts the connection at a right angle, regardless of where you click.
 - **Connector+Alt**: *Plant Simulation* sets anchor points of the connection at the location where you click. This is similar to the freehand drawing feature.
- To create a **non-straight connection**, set one or more anchor points: First, click object A, proceed to point 1 and click once, then proceed to point 2, click once, and finally move to object B and click the mouse button once to establish the connection.



- To create a **right-angled connection**, hold down **Shift** and click the left mouse button to set the anchor point.
- To move an anchor point, click it, hold the mouse button down, and drag the handle  to the desired location. *Plant Simulation* also moves the *Connector*.
- To check if all connections between the objects really have been established, click **Unconnected Objects** on the **View** ribbon tab. *Plant Simulation* selects all objects with unconnected entrance and exit points.
- To make the connecting line thicker, enter a higher number into the text box **Weight**.
To make the connecting line thinner, enter a lower number into the text box **Weight**.
- To select another color than the default black, in the dialog **Colors**, click the field next to **Color**.

You can select one of the predefined colors, or you can click **More Colors** and click the **Select** button to select a color in the color matrix. Then click **OK**. *Plant Simulation* shows this color next to **More Colors** and uses it as the active color.

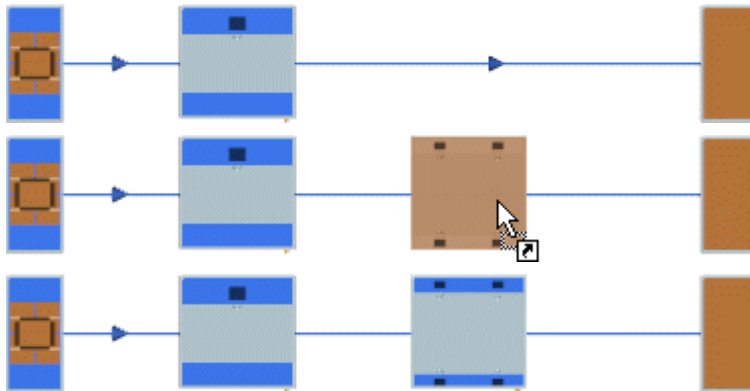
- To show the source and the target of the *Connector* in the *Frame* is a *Tooltip*, drag the mouse over the *Connector*.



- To **exchange the successor** of a *Connector*, select the end point of the *Connector* and drag it to another object.

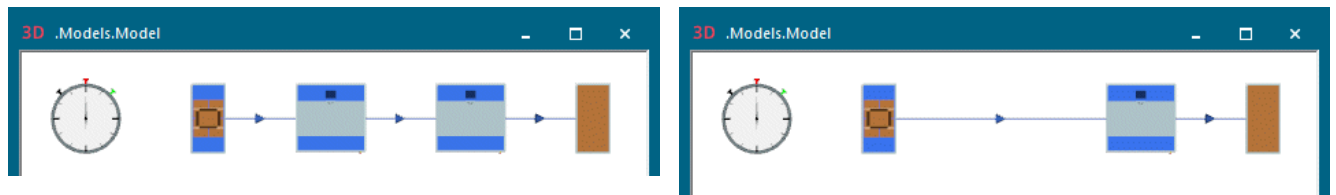
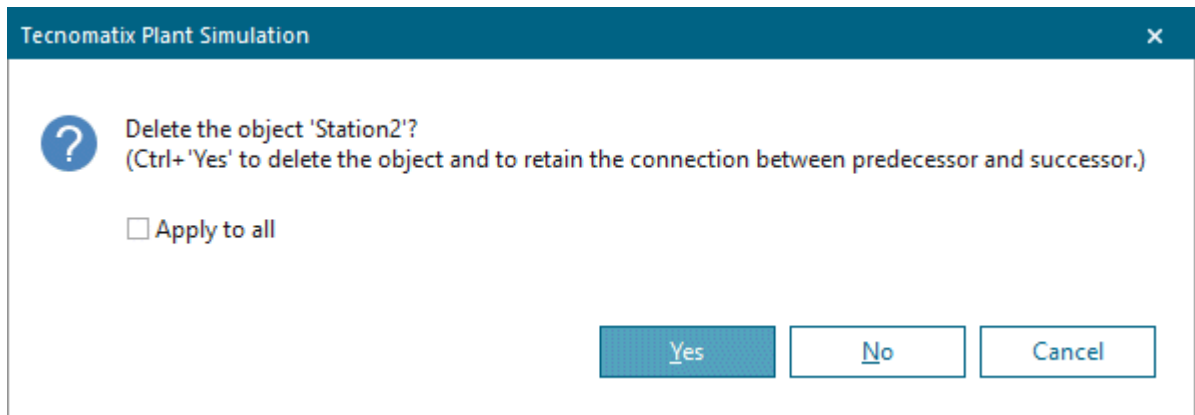
To **exchange the predecessor** of a *Connector*, select the starting point of the *Connector* and drag it to another object.

- To **place an object**, which you insert from the *Class Library* into a *Frame*, **in between two already connected objects**, drag that object onto that spot of the *Connector* and drop it there. In this process *Plant Simulation* exchanges the successor of the original *Connector*.

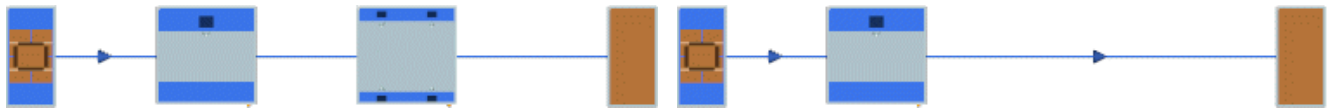


This also works when you drag a *Frame*, which contains exactly one suitable **entrance Interface** and exactly one suitable **exit Interface**, onto the *Connector*.

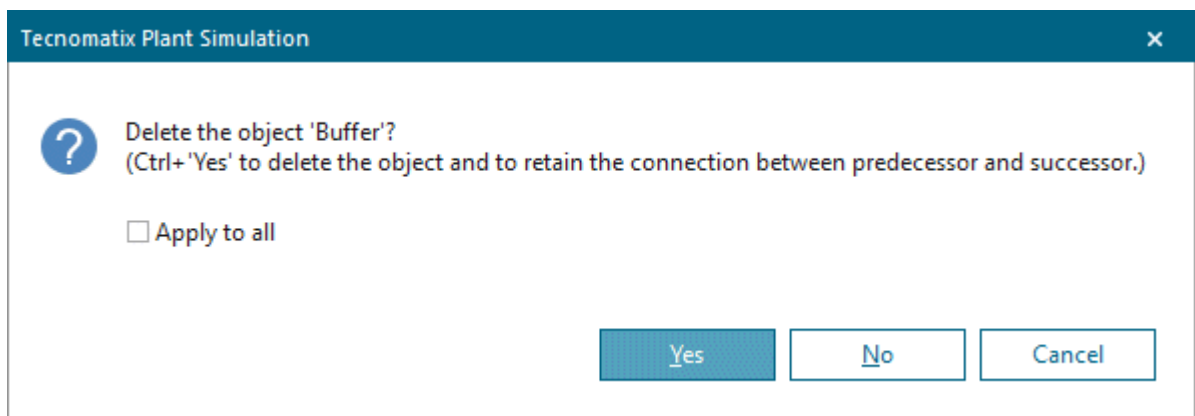
- To retain the *Connector* between objects when you delete an object, hold down **Ctrl** and then press **Del** while keeping **Ctrl** pressed. Click **Yes** in the dialog that opens.



- To **automatically connect predecessor and successor when deleting an object** located between other connected objects, hold down **Ctrl** while you delete the object with the context menu command **Delete**.



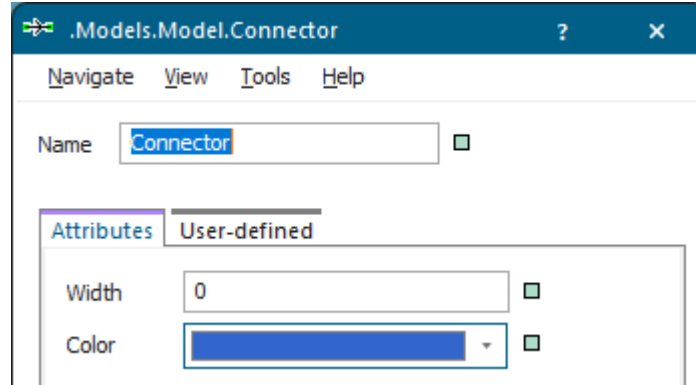
When you work with the **Delete** key, select the object(s) you want to delete, and press **Delete**. Follow the instructions in the dialog that opens.



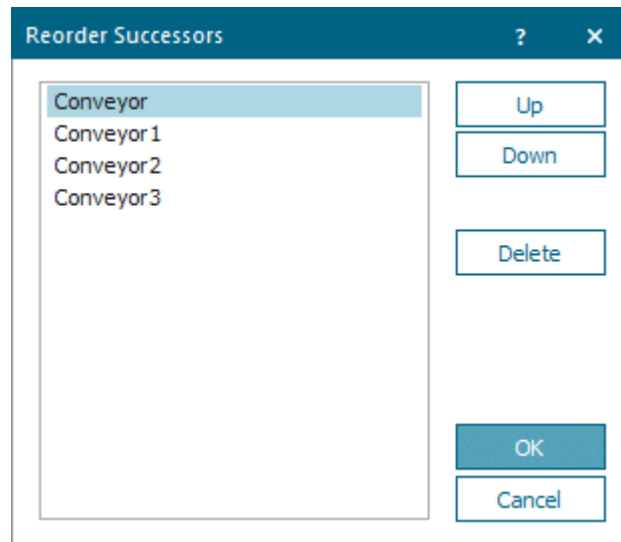
Note

This does not work when an object has more than one predecessor and successor. If the object has a successor, *Plant Simulation* exchanges the successor of all preceding *Connectors*. If the object has a predecessor, *Plant Simulation* exchanges the predecessor of all succeeding *Connectors*.

- To change the **Width** and the **Color** of the *Connector*, double-click its icon and enter and select the settings you need.



- To **reorder the sequence of the succeeding objects**, i.e., the sequence of the *Connectors*, click the object with the right mouse button and select **Reorder Successors**. Move the selected successor with the **Up** or **Down** buttons.



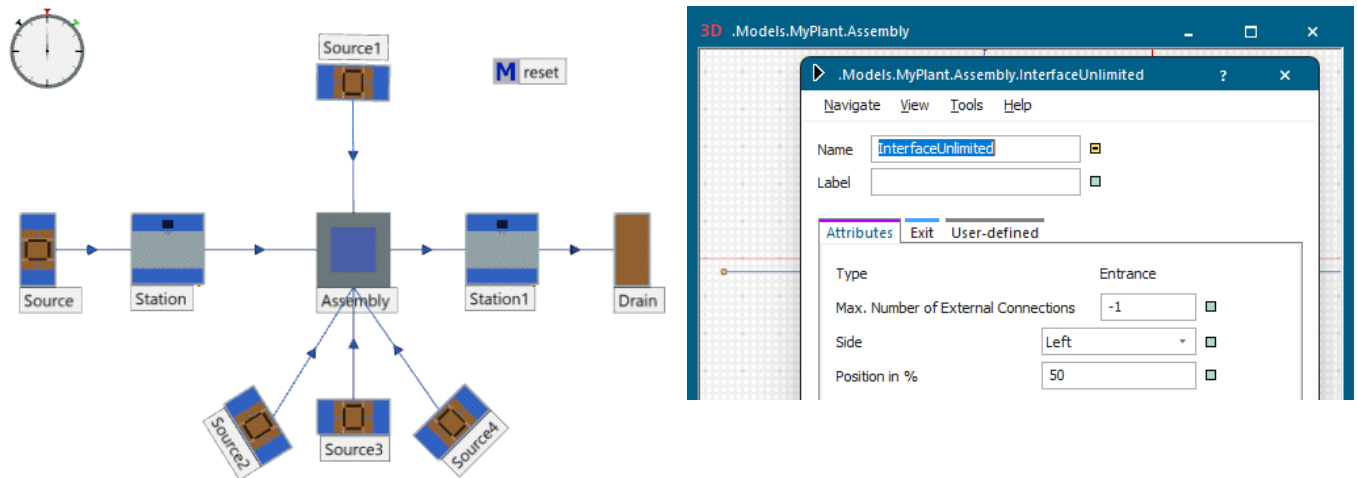
Compare [Auto-connect Length-oriented Objects](#)

Compare [Undo \[3D\]](#) and [Redo \[3D\]](#)

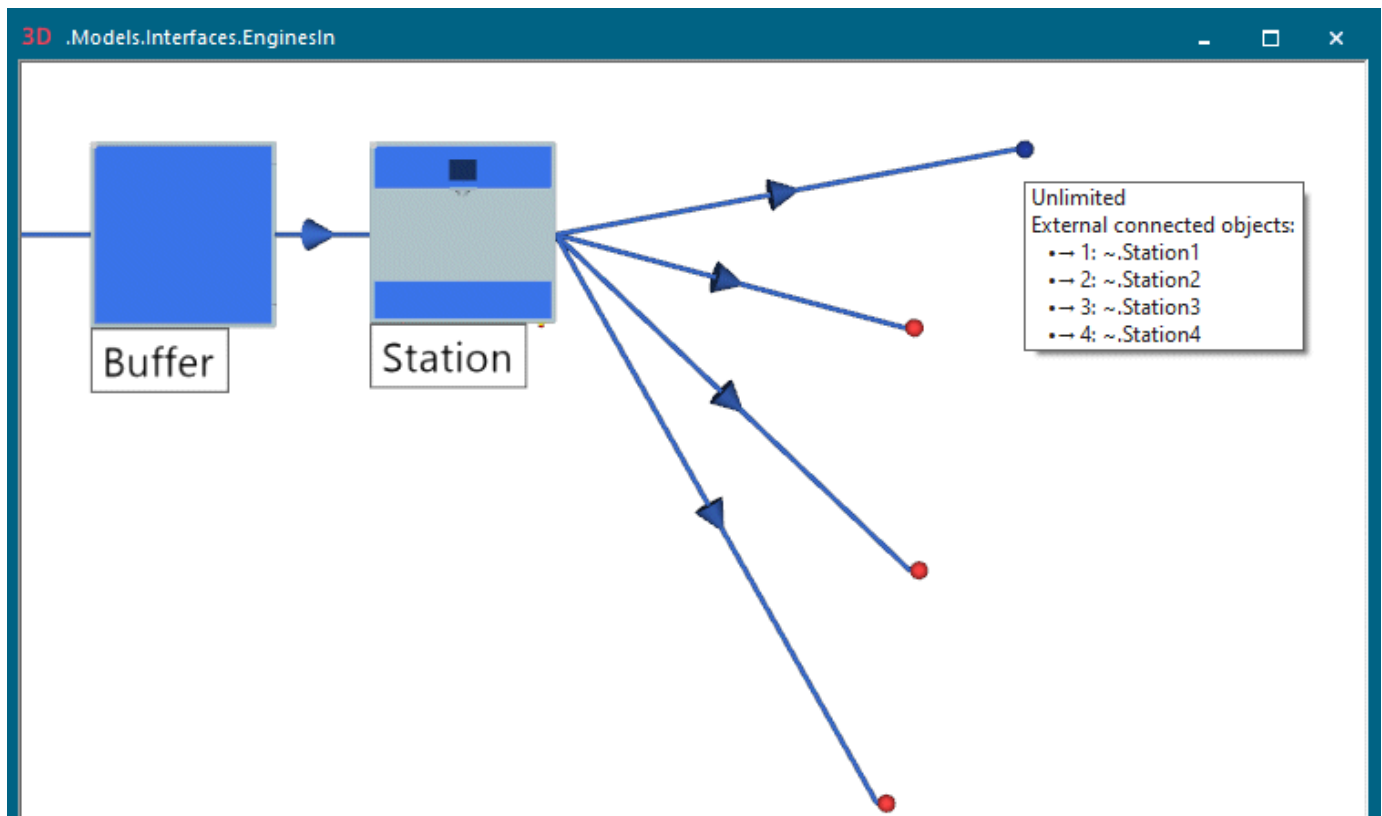
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Model Transitions Between Frames

One of the main advantages of *Plant Simulation* is that you can model hierarchically, i.e., you can model components of your overall plant in a *Frame*, make sure that these components work by themselves and then insert them into your overall simulation model.

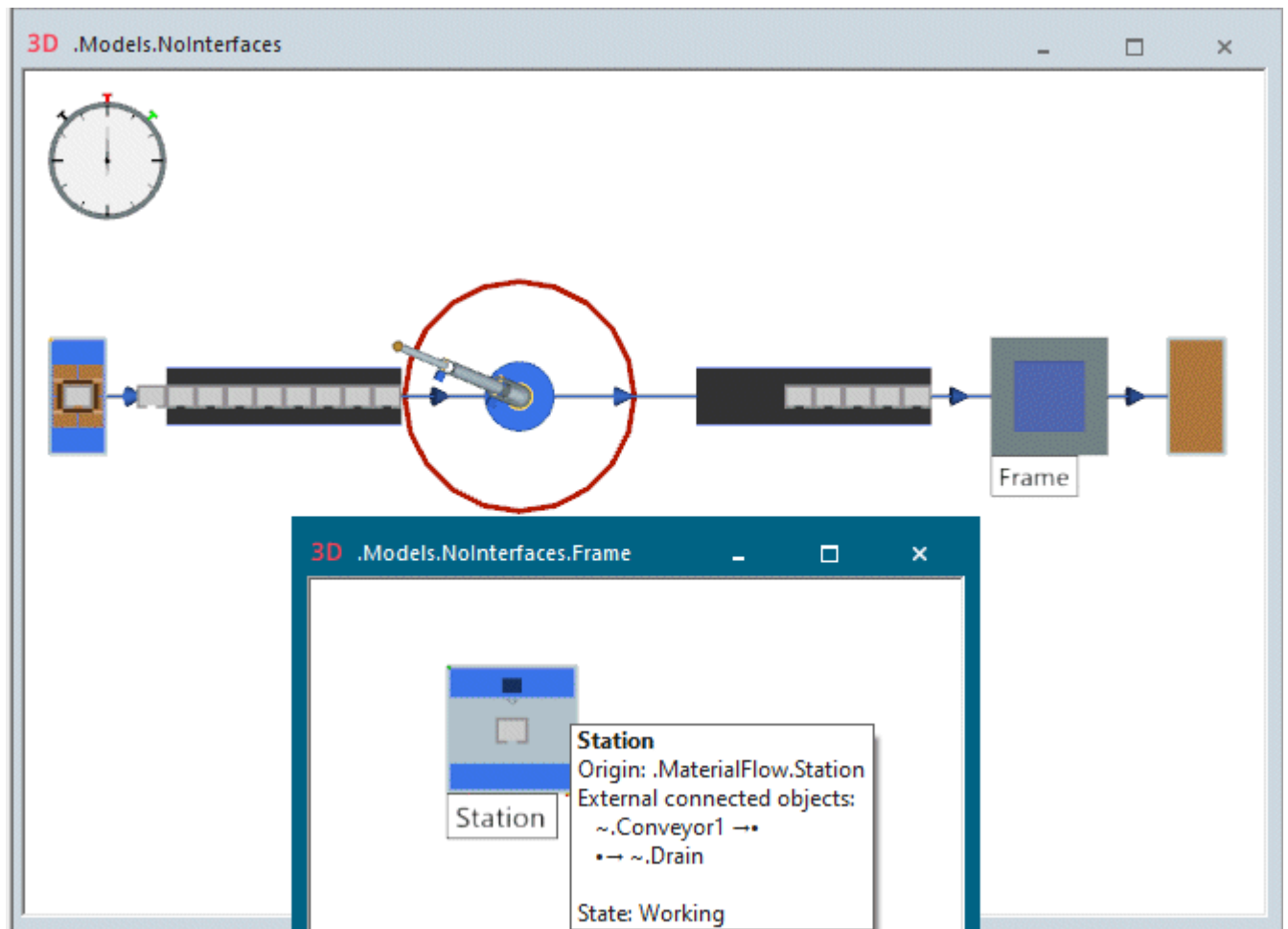


In this case, you will have to connect the component (*Frame*) to the material flow objects or *Frames* that precede and succeed it in the sequence of stations with the object **Interface** ▶. The transitions are the places where *MUs* move from one *Frame* to another or from a material flow object to a *Frame* and vice versa. *Plant Simulation* shows if the *Interface* you inserted into a *Frame* is connected to another object with a *Connector* or if it is not connected. *Plant Simulation* also recognizes if an *Interface* is an **entrance** or an **exit**.

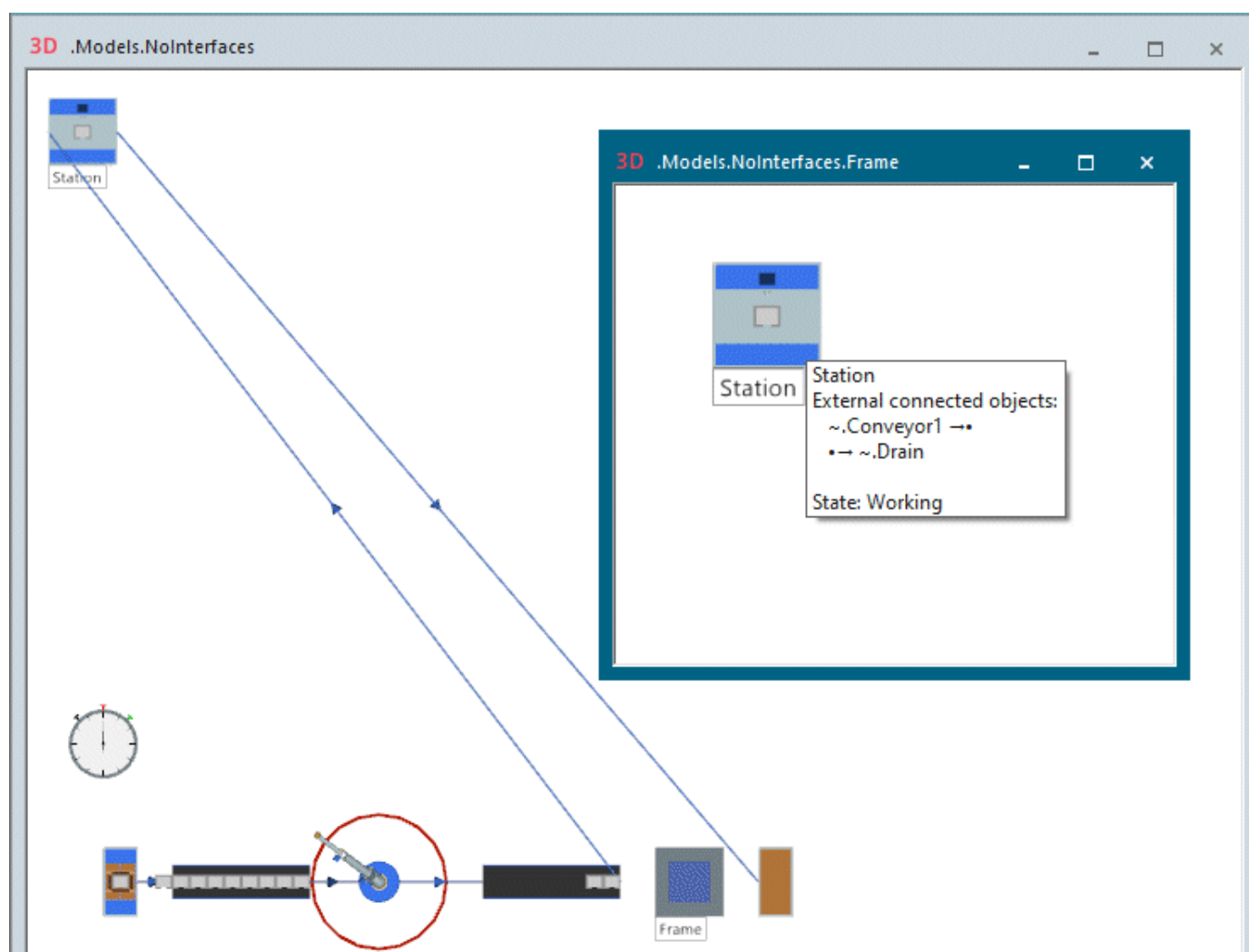


If the *sub-Frame* only contains a **single material flow object**, you do not have to use *Interfaces* but can connect the *sub-Frame* directly with its predecessor and its successor.

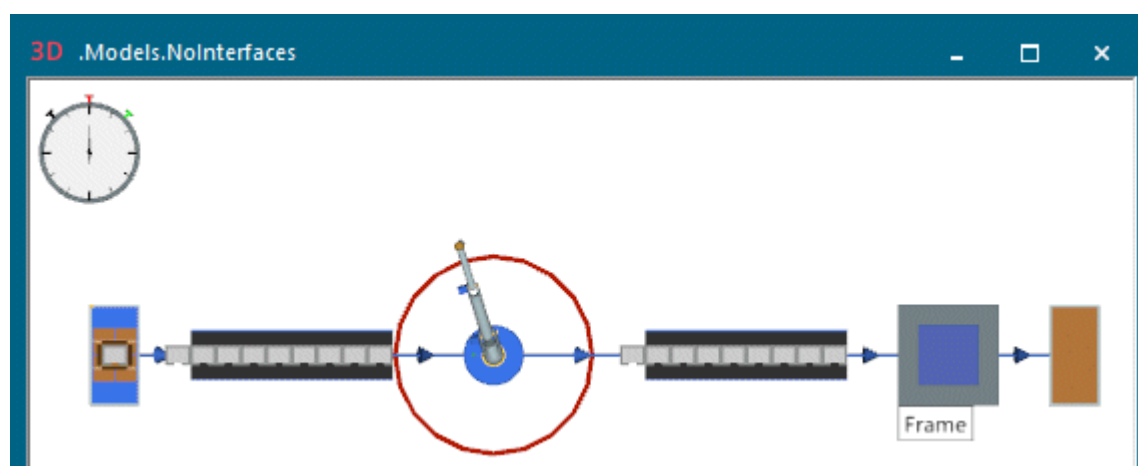
The *Tooltip* shows the externally connected objects. *Plant Simulation* does not insert a *Connector* from the predecessor and to the successor of the object in the *sub-Frame*.



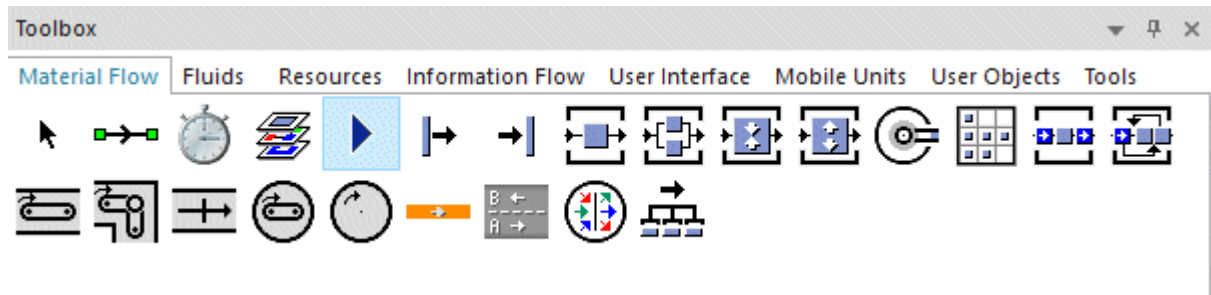
Our model looks like this in 3D by default as **Graphics > Show Content** is activated by default:



To only show the *sub-Frame*, clear **Show Content** for the *Frame* in the model **NoInterfaces**. The model then looks like this:



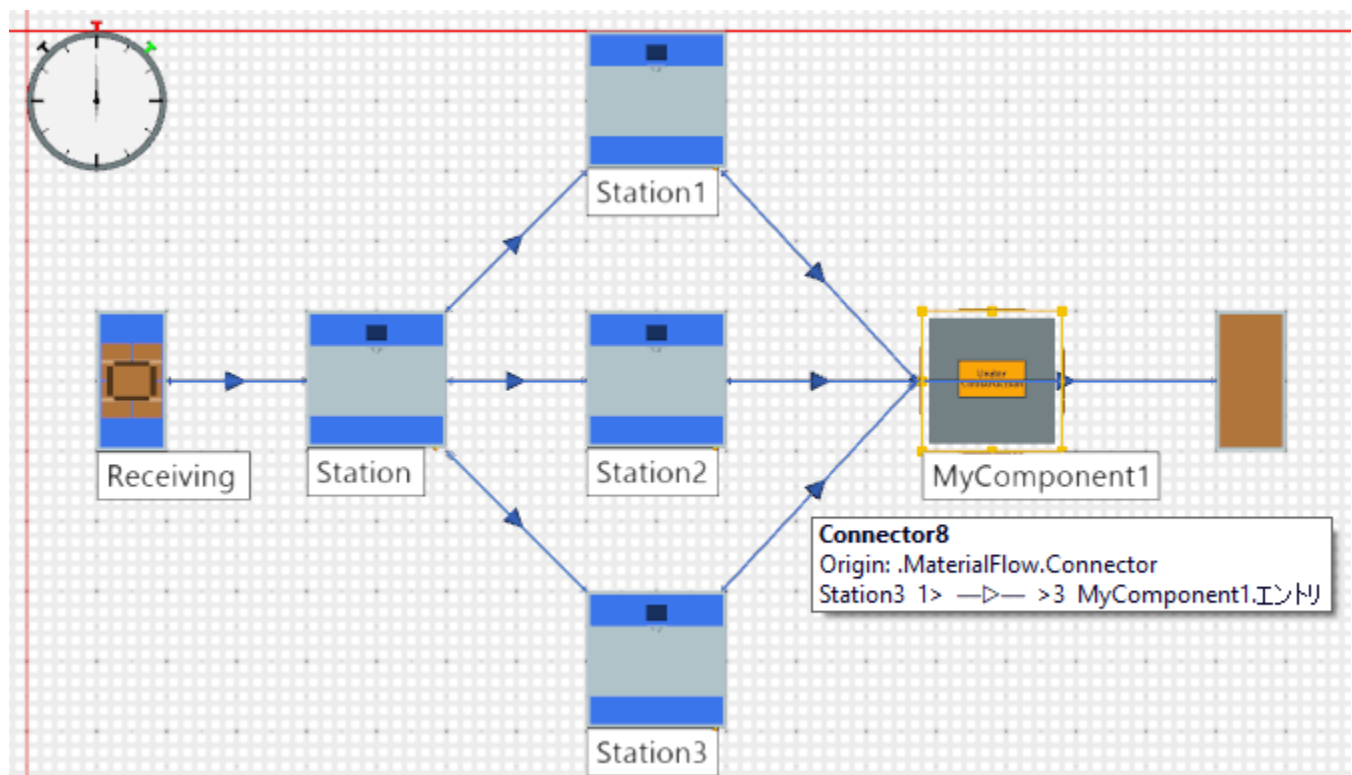
You can insert the *Interface* into your simulation model from the folder **MaterialFlow** in the *Class Library* or from the toolbar **Material Flow** in the *Toolbox*.



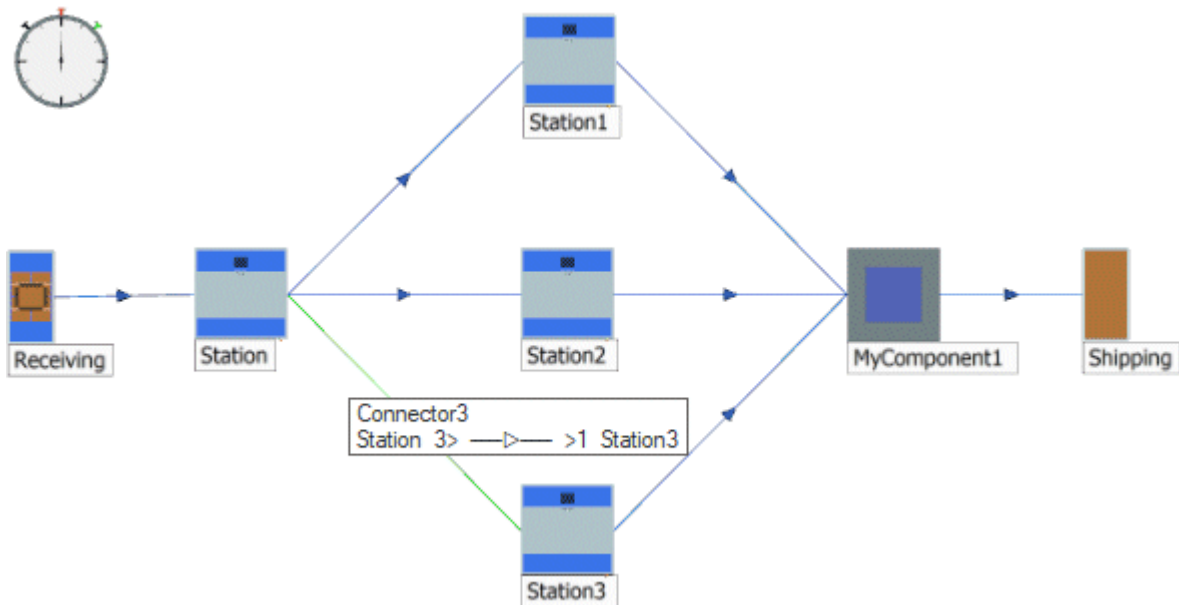
To model transitions, click the folder or the toolbar **Material Flow** and insert as many objects of type *Interface* anywhere in the *Frame*.

- Select the side of the icon of the *Frame* at which the *Interface* is to be located: on the **Top**, on the **Right**, on the **Bottom**, or on the **Left** of the *Frame*.
- Type in the maximum number of external connections the *Interface* may have. Depending on the type, any number of *Interfaces* may have more than one predecessor or successor.

The **predecessor** is the object that is connected to the selected object with a *Connector* and that is located in front of it in the sequence of stations in the simulation model.



The **successor** is the object that is connected to the selected object with a *Connector* and that is located after it in the sequence of stations in the simulation model.



- Type the **position** of the *Interface* on the side of the *Frame* it is inserted into in percent into the text box **Position in %**. You can specify a value between 0 and 100 percent. *Plant Simulation* uses the value you specify, when you activate **File > Preferences > General > Connect Objects Automatically**. Connecting objects automatically only works when the exit of *FrameA* and the entrance of *FrameB* are not more than three pixels apart.
- An *Interface* object you insert into a *Frame* and connect with a *Connector* show its **Type** in the dialog: either **Entrance** or **Exit**.
- If need be, you can also select an **Exit Strategy** on the tab **Exit**.

Compare [Model Hierarchically \[general\]](#)

Compare [Transfer Parts from Station to Station](#)

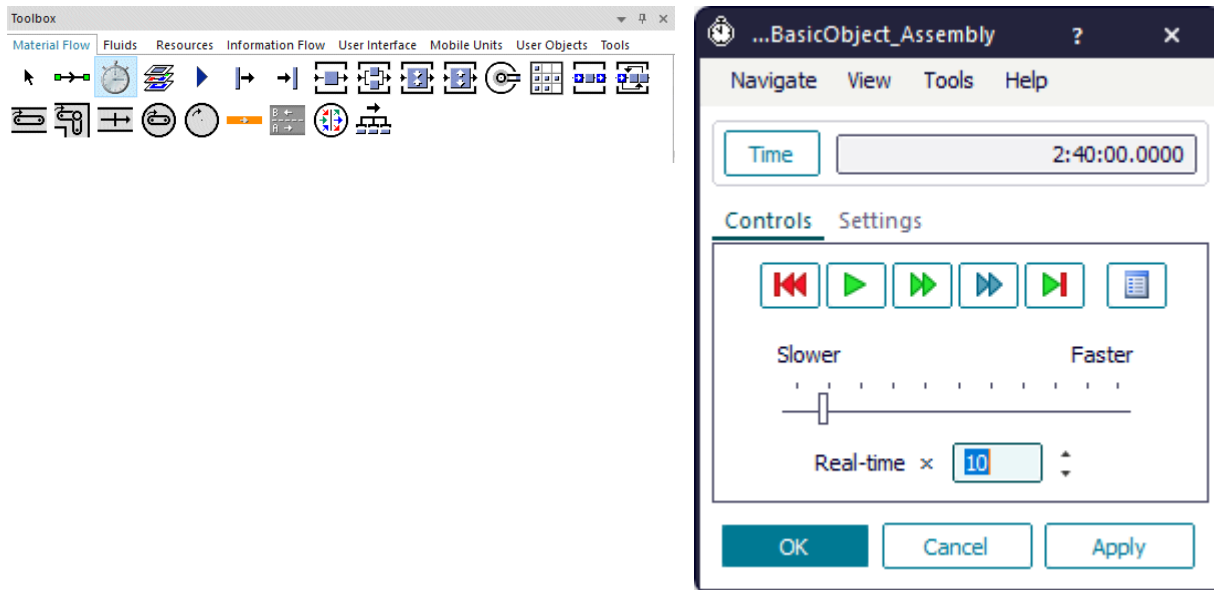
Back to [Working with the Frame](#)

Controlling the Simulation with the EventController

Control the Simulation with the EventController

The *EventController* coordinates and synchronizes the different events taking place during a simulation run. When a *part* enters a processing station, for example a *Station*, *Plant Simulation* computes the time it takes to process it and enters that event and that time into the **List of Scheduled Events** of the *EventController*.

You can insert the [EventController](#)  into your simulation model from the folder **MaterialFlow** in the *Class Library*, from the tab **Material Flow** in the *Toolbox*. Instead, you can also click  on the **Home** ribbon tab.



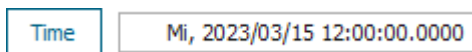
- *Plant Simulation* show the current simulation time of the simulation run in the box next to **Time**. Click **Time** to change the display of the time between the:
 - **Relative time:** *Plant Simulation* resets the relative time to zero, when it starts the simulation run. This is the default.
 - **Current time plus simulation time:** *Plant Simulation* adds the simulation time to the time and date at which it started the simulation run.

Let us assume today is **March 13, 12 o'clock noon** and the simulation is to run for two days:

- For the setting **relative time shows 2:00:00:0000**, i.e., two days.



- For the setting **current time plus simulation time** *Plant Simulation* shows **We, 2023/03/15 12:00:00.0000**, i.e., March 15, 12 o'clock noon, after the simulation run is finished.



- To initialize your simulation model, click **Start/Stop Simulation** .

Plant Simulation executes all *Methods* named *Init*. *Methods* being executed when you click **Start/Stop Simulation** will be processed first.

Plant Simulation initializes the simulation model before it executes the next scheduled event.

Note

Plant Simulation automatically executes *init methods* then you start the simulation for the first time or when you stopped the simulation and clicked **Reset Simulation** and you did not click **Start/Stop Simulation** before you started the simulation.

- To reset your simulation model, do one of the following:

- Click **Reset Simulation**  in the *EventController*.

- Click  on the **Home** ribbon tab.

Plant Simulation calls all *methods* named *Reset* in your simulation model. It deletes all unprocessed events, resets the simulation time to 0, resets the statistics, and clears any pause and any failure of paused or failed machines.

When you click **Reset Simulation** while the simulation is running, *Plant Simulation* will first finish processing the active event and then reset the simulation model.

- To start the simulation, do one of the following:

- Click **Start/Stop Simulation**  in the *EventController*.

- Click  on the **Home** ribbon tab.

- Double-click the icon of the *EventController* while holding down **Shift**.

If this is the first simulation run or you clicked **Reset Simulation** before, *Plant Simulation* executes all *methods* named *Init* located in your simulation model.

Note

To start the simulation without animating objects and *MUs*, click **Start Fast Forward Simulation**  on the **Home** ribbon tab.

- To stop the simulation after the active simulation event has been processed, do one of the following:

- Click **Start/Stop Simulation**  in the *EventController* again.

- Click  on the **Home** ribbon tab.

- Double-click the icon of the *EventController* while holding down **Shift**.

- To start the simulation run with the animation of **MUs and States**  and with full speed, click **Start Full Speed Simulation** .

- To start the simulation run without the animation of **MUs and States**  and with maximum speed, click **Start Fast Forward Simulation** . This makes the simulation run as fast as possible.

- To process the next simulation event in line, to stop the simulation afterward, and to thus proceed through your model step-by-step, click **Single Step Simulation** .

- To open the **Dialog Box of the Event Debugger**, click **List** .

The **List** of scheduled events shows all currently scheduled events in ascending order, allowing you to find out, which events are scheduled at which point in time.

- The column **Breakpoint** shows an **S**, for a breakpoint you inserted, by double-clicking a cell in the row.
- The column **Type** shows the type of event, for example **Out**, **Pause**, **PauseEnd**, etc.
- The column **Time** shows the point in time the event is going to be executed.
- The column **Receiver** shows the receiving object, i.e., the object that receives the breakpoint.
- The column **Sender** shows the sending object, i.e., the object that sends the breakpoint.
- To increase the speed of the simulation, drag the slider to the left or press the left arrow key. To decrease the speed of the simulation, drag it to the right or press the right arrow key.

Decreasing the speed enables you to better follow the movements of *MUs*, as *Plant Simulation* show their icons for a longer time on the same station.

- To pause the simulation until the time span between two events has elapsed in real-time, enter the **Scaling Factor** for real-time mode.

The scaling factor sets the time that elapses between two events in real-time.

Note

Normally *Plant Simulation* does not take the time span in between events into account, as it is a discrete event simulation system.

The duration of an event in real-time is the simulation time divided by the scaling factor you entered. The resulting duration is an integer.

You can:

- [Select Settings for the Simulation](#)
- [Work with the Event Debugger](#)

Back to [Creating a Simulation Model](#)

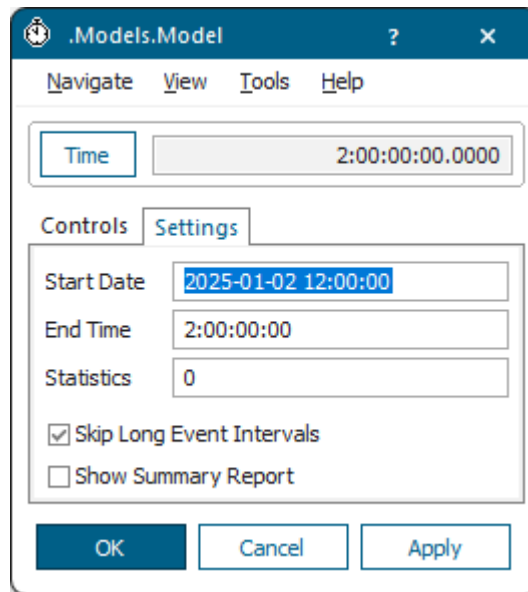
Select Settings for the Simulation

When your modeling needs require it, you can select settings for controlling the simulation run on the tab **Settings** of the *EventController*.

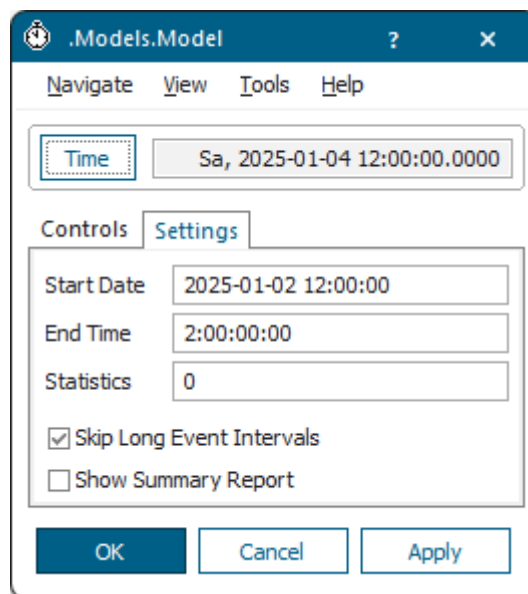
- *Plant Simulation* show the current simulation time of the simulation run in the box next to Time.
Click **Time** to change the display of the time between the:
 - **Relative time:** *Plant Simulation* resets the relative time to zero, when it starts the simulation run. This is the default.
 - **Current time plus simulation time:** *Plant Simulation* adds the simulation time to the time and date at which it started the simulation run.

Let us assume today is the second of January, 12 o'clock noon and the simulation is to run for two days:

- For the setting **relative time** *Plant Simulation* shows **2:00:00:0000**, i.e., two days.



- For the setting **current time plus simulation time** *Plant Simulation* shows **Mi, 2023/01/04 12:00:00.00**, i.e., January 04, 12 o'clock noon, after the simulation run is finished.



- Enter the **Date** and the **time** on which the absolute time during the simulation is based.
- Enter the **Time** when the simulation run will be finished. Enter a relative time, i.e., the period of time the simulation runs. *Plant Simulation* compares this period with the simulation time and stops the simulation run when both are identical. Let us assume it is January 04, 12 o'clock noon and the simulation is to run for two days. Enter **2 : 00 : 00 : 00** into the text box **End**. If you do not want to write this out in full, you can also just type **2 : : :** and click **Apply** to have *Plant Simulation* translate this to the full format **2 : 00 : 00 : 00**.
- Enter the **Time** the *EventController* resets the statistics. *Plant Simulation* starts collecting statistical data for all material flow objects anew from this time on.

Back to [Control the Simulation with the EventController](#)

Working with the Event Debugger

Work with the Event Debugger

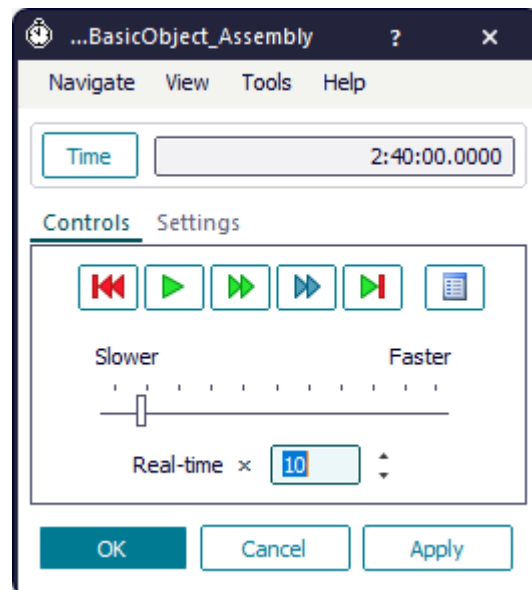
The *Event Debugger* is a tool that lets you accurately control the execution of simulation events, when you click **Single Step Simulation**. This way you can follow how *Plant Simulation* processes the individual events during the simulation run and detect errors in your simulation model.

As opposed to clicking **Start/Stop Simulation** on the tab **Controls** of the *EventController* you can, for example, enter a condition when the *EventDebugger* is to **Stop** the simulation. Then, you can follow the execution of the events in the *Event Debugger*.

The following examples demonstrate a number of typical usages for the *Event Debugger*.

To open the *Event Debugger*, to define breakpoints and to step through the scheduled events:

- Open the *EventController*.

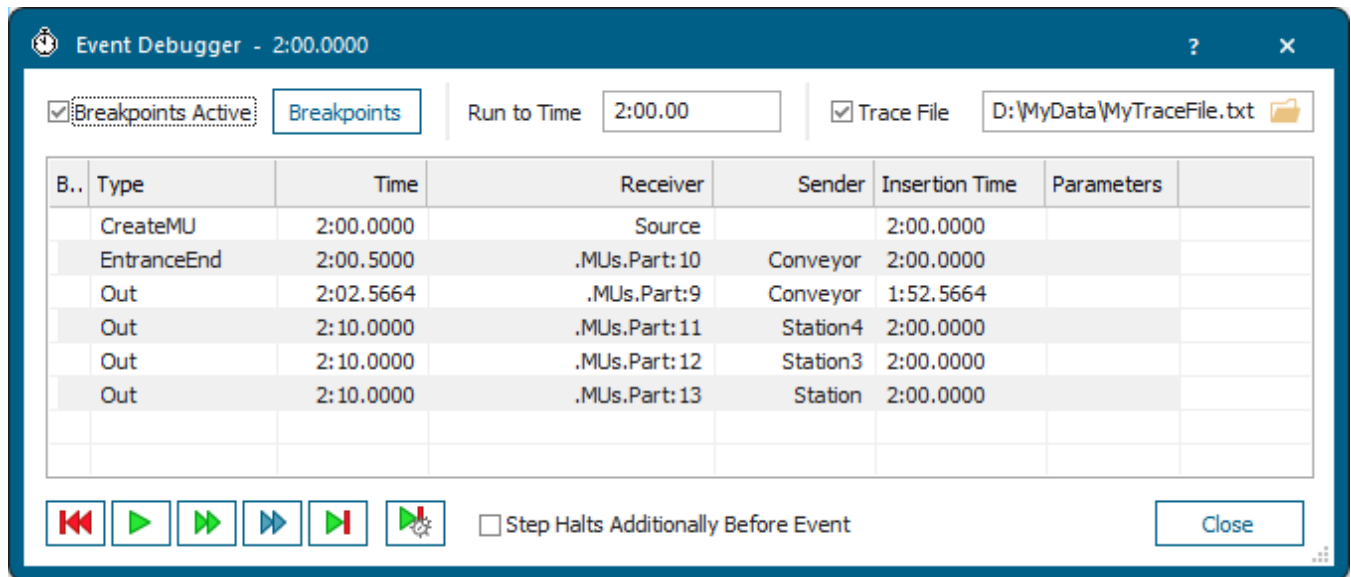


- Click the **List** button  on the tab **Controls**. The event list contains the types of scheduled events, which the objects in your simulation model entered and which the *EventController* has to process. It is sorted ascending by time, tells the type of event, the scheduled processing time, and lists recipient and sender of the event.

Double-click a cell in the row where you would like to insert a single breakpoint. Then, the *Event Debugger* stops the simulation run immediately before the *EventController* processes that event. Double-click the row again to delete the breakpoint.

- Select the check box **Breakpoints Active**.
- Click **Breakpoints** to open the list of the breakpoints you defined.

- Click **Insert** in the dialog **Breakpoint** to add an additional breakpoint.



- If you would like *Plant Simulation* to create a trace file that tracks **all events**, enter a name into the text box **Trace File** and select the check box **Trace File**.

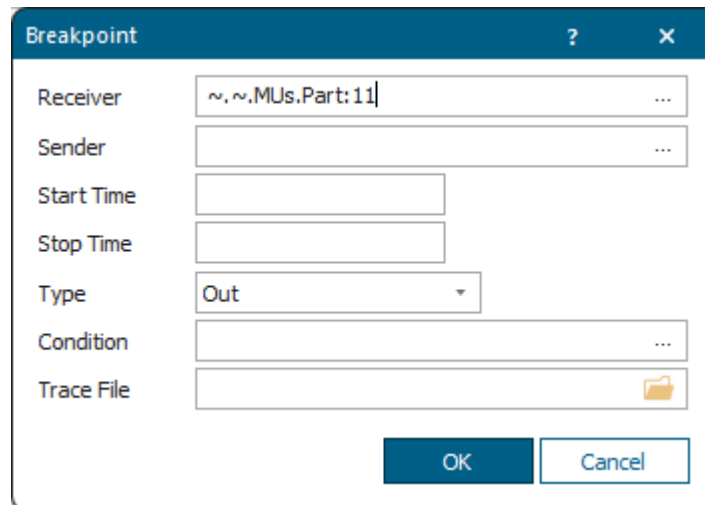
If you only want to create a trace file for a **certain event**, enter a file name into the text box **Trace File** in the dialog **Breakpoint**.

- You can also use an event from the list in the *EventController* to define breakpoints. Select the event in the list and click **Stop at Selected Event**. This adds this event to the list of defined event breakpoints. You can also edit the selected event.
- To process a single event and stop the simulation again, click **Single Step Simulation** .
- To proceed with the simulation until the *EventController* reaches the next breakpoint, click **Start/Stop Simulation**.

Back to [Work with the Event Debugger](#)

EventDebugger, Example 1

In this example we watch all exit (**Out**) events for the *Part* with the ID 1, meaning that we track the course the *Part* takes through the plant.



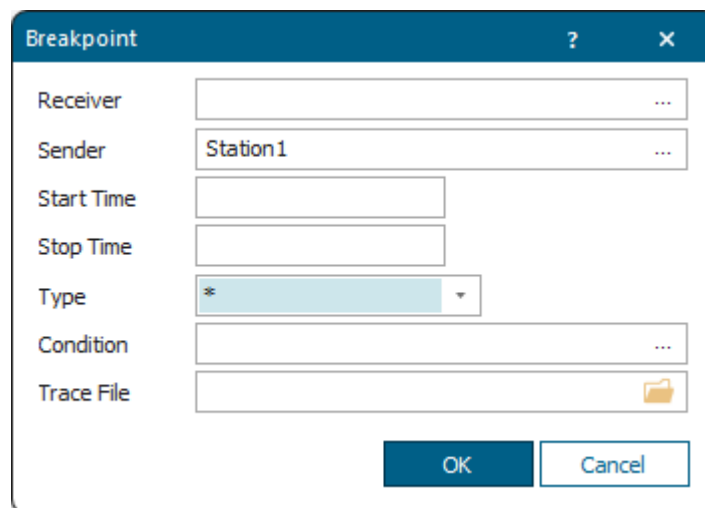
Breakpoint	
Receiver	~.~.MUs.Part:11
Sender	
Start Time	
Stop Time	
Type	Out
Condition	
Trace File	
OK Cancel	

If you would like to create a breakpoint for each *Part*, delete the ID of the *MU*.

Back to [Work with the Event Debugger](#)

EventDebugger, Example 2

In this example all *MUs* leaving *Station1* define a breakpoint.



Breakpoint	
Receiver	
Sender	Station1
Start Time	
Stop Time	
Type	*
Condition	
Trace File	
OK Cancel	

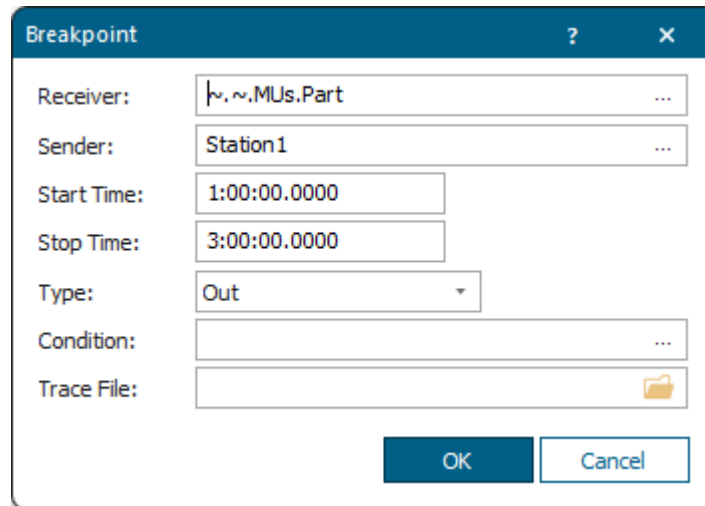
This time we do not enter a **Receiver**, as we want the *EventController* to insert a breakpoint for each leaving *part*. The **Sender** is *Station1*.

Back to [Work with the Event Debugger](#)

EventDebugger, Example 3

In this example only *MUs* of a certain class located on *Station1* insert a breakpoint during a certain time span.

Plant Simulation inserts a breakpoint when a *part* of the class `.MUS.Part` located on *Station1* creates an **Out** event between 1 hour and 3 hours.



Breakpoint	
Receiver:	<code>.MUS.Part</code>
Sender:	<code>Station1</code>
Start Time:	1:00:00.0000
Stop Time:	3:00:00.0000
Type:	Out
Condition:	
Trace File:	
OK Cancel	

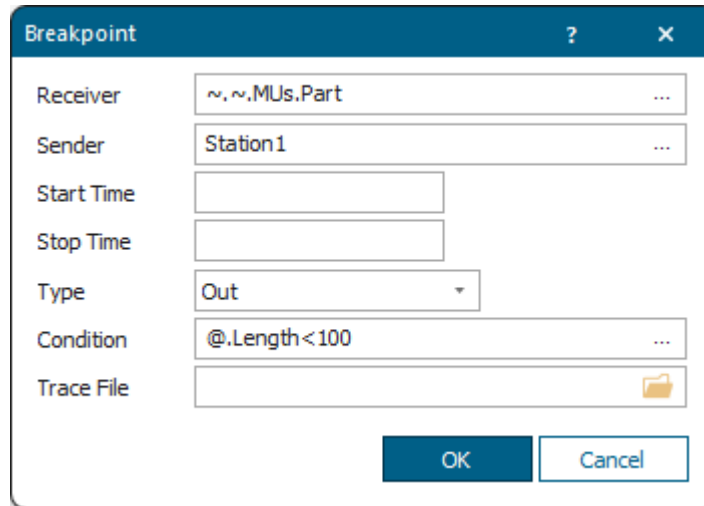
At times you only want to insert breakpoints when certain conditions are met. Enter these conditions into the text box next to **Condition**. A condition may be a certain value or certain states of the simulation model or properties, i.e., attributes, of mobile parts. You can also enter methods, returning a boolean value as a result, which will then be evaluated.

Back to [Work with the Event Debugger](#)

EventDebugger Example 4

In this example we use a property of the *part* as a condition for a breakpoint.

Plant Simulation inserts a breakpoint, when a *part* of the class `.MUS.Part`, which is less than 100 meters long, causes an **Out** event on *Station1*. The unit depends on the settings you selected under **File > Model Settings/Preferences > Units > Length**.



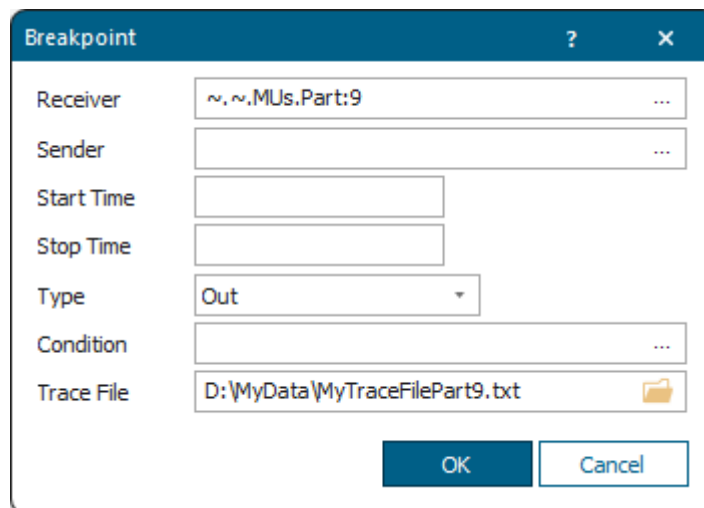
Breakpoint	
Receiver	~,~,MUs.Part
Sender	Station1
Start Time	
Stop Time	
Type	Out
Condition	@.Length<100
Trace File	

OK Cancel

Back to [Work with the Event Debugger](#)

EventDebugger, Example 5

In this example we track the course the *Part* with the **ID 9** takes through a simulation model and write the stations to a trace file.



Breakpoint	
Receiver	~,~,MUs.Part:9
Sender	
Start Time	
Stop Time	
Type	Out
Condition	
Trace File	D:\MyData\MyTraceFilePart9.txt

OK Cancel

With the settings shown in the dialog above, *Plant Simulation* inserts a breakpoint for each **Out** event of the *Part* with the **ID 9** and writes this event to the file `c:\Exercises\trace3.txt`. This only works if you selected the check box **Trace File** in the dialog **Event Debugger**. If you clear the check box **Breakpoints Active**, *Plant Simulation* writes the trace file without stopping your simulation using a breakpoint.

In the examples above we only used **Out** events. You can, of course, also use any of the available types of events, compare the [List of Events](#).

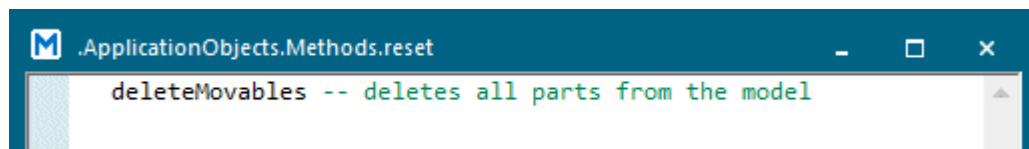
Back to [Work with the Event Debugger](#)

Delete Parts with the Mouse or when Resetting the Model

You can delete parts with the mouse or when resetting the simulation model.

To delete one, several or all *MUs*, do one of the following:

- Select a single *MU* you would like to delete and press **Delete**.
- To delete all *MUs* in all objects in the active *Frame* and its *sub-Frames*, click  **Delete MUs** on the **Home** ribbon tab.
- To delete all mobile objects (*MUs*), when you reset the simulation model by clicking the **Reset Simulation** button or by calling the method `reset`, you can enter `deleteMovables` into a *Method*, which you name `reset` and which you insert into your simulation model.



Note

If your simulation model still contains parts, meaning that *Plant Simulation* does not start counting parts with 1, for example `.MUs.Part:1`, for a new simulation run, check if the model contains additional *Frames* with inserted *EventControllers*. Reset the *EventControllers* there as well.

Back to [Creating a Simulation Model](#)

Working with Drag-and-Drop

Work with Drag-and-Drop

Drag-and-drop has a number of applications in *Plant Simulation*.

You can:

- [Drag an Object Onto Another Object and Drop It There](#)
- [Use a Drag-and-Drop Control for Several Objects](#)

Compare [Object Paths in SimTalk](#)

Compare [Drag-and-Drop in the Icon Editor](#)

Compare [Drag-and-Drop While Modeling](#)

Compare [Drag-and-Drop in the Method Editor](#)

Compare [Drag-and-Drop in Lists and Tables](#)

Compare [Work with Data in the DataTable](#)

Compare [Drag-and-Drop in the Chart](#)

Compare [Configuring the TransferStation](#)

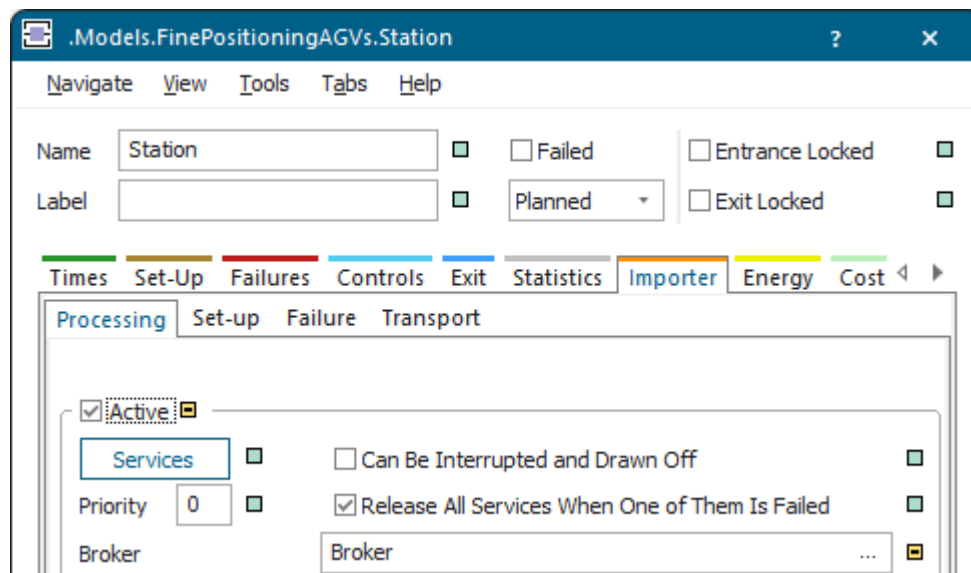
Back to [Creating a Simulation Model](#)

Drag an Object Onto Another Object and Drop It There

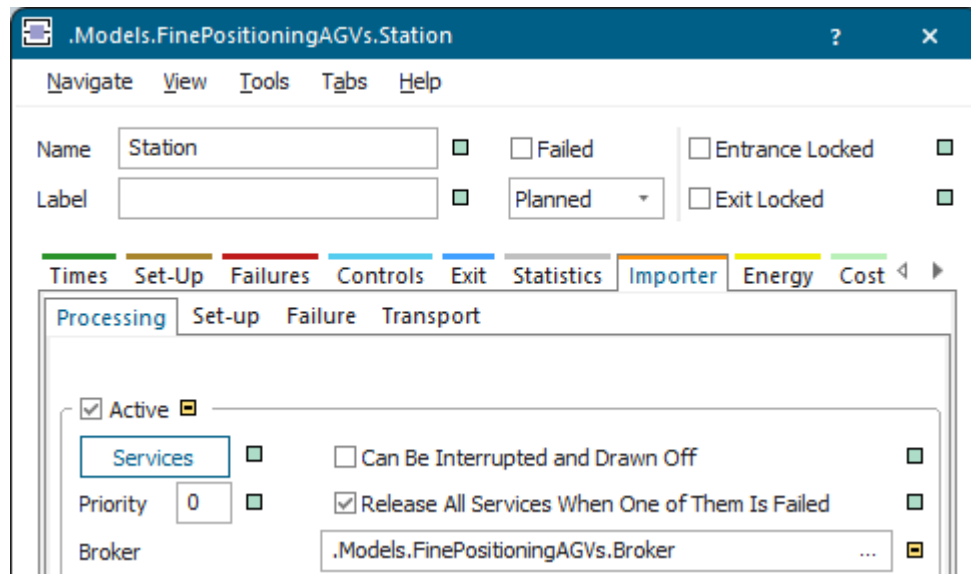
Drag an object onto another object and drop it there.

Drag-and-drop has a number of applications in *Plant Simulation*. You can use it to:

- Insert an object from the *Class Library* into the *Frame* that contains your simulation model.
- Insert an object, such as a *Broker*, a *ShiftCalendar*, a *control method*, a *table*, etc. into the text boxes the dialogs of the objects provide. When you use drag-and-drop, *Plant Simulation* enters the **relative path** to the object. It starts in the *Frame* in which the object using the relative path is located.



To insert the absolute path of the object, hold down the **Shift** and **Ctrl** keys.

**Note**

To use an **object reference** instead of the absolute path, type an asterisk in front of the path. For the example above you would type `*.Models.FinePositioningAGV.Broker`.

Note

If you do not want to use the absolute path, but the **relative path**, click ... **Select Object**.

- Duplicate i.e., to copy the selected object or model and create a new class: Hold down the **Ctrl** key, drag the object to another location in the *Class Library*, and release **Ctrl**.

Go to [Use a Drag-and-Drop Control for Several Objects](#)

Back to [Work with Drag-and-Drop](#)

Use a Drag-and-Drop Control for Several Objects

Plant Simulation supports **multiple drag-and-drop**, allowing you to select a number of objects, drag them onto the destination object and drop them there.

You can do this for:

- Each object which has a **Drag-and-drop control** which accepts the selected objects as a parameter of data type *array*.
- The [AttributeExplorer](#)
- The [Method](#). Here the selected objects to be passed have to be contained in an *array*.
- The [Chart](#)
- The [HtmlReport](#)

- The object **Cycle**. It accepts a maximum of two objects, one for the first station and one for the last station.
- The **ShiftCalendar**
- The **LockoutZone**

You can also drag several objects to the structure in the *Class Library* to create class objects from the copied objects.

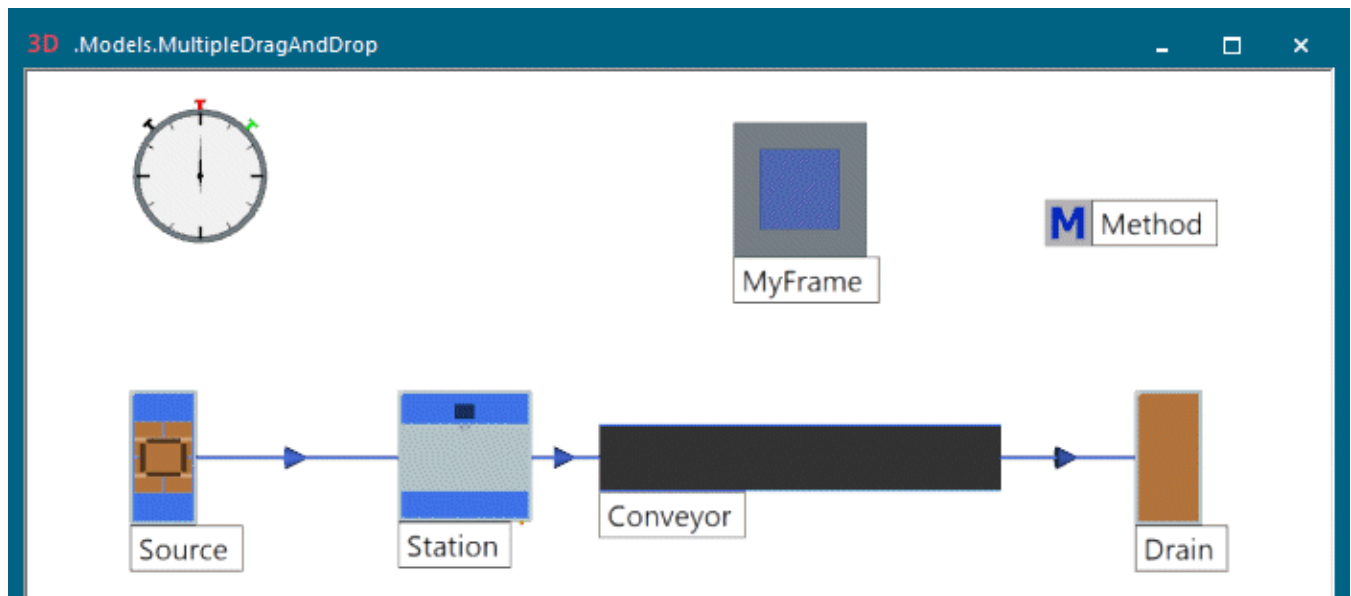
Below we demonstrate how to use **multiple drag-and-drop** with a **drag-and-drop control**.

We want to execute the **drag-and-drop control**, which we entered for the *Frame* named *MyFrame*, when we drag our production line onto the *Frame* and drop it there.

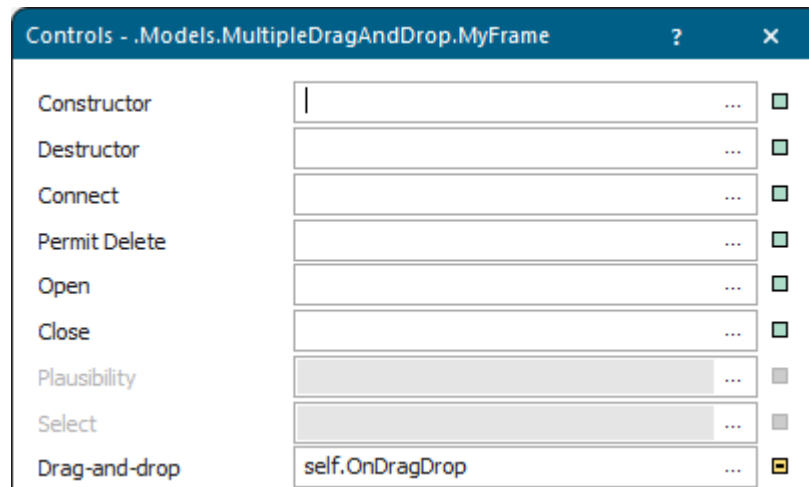
Proceed as follows to create a simple simulation model:

- Insert the material flow objects you want to work with and connect them. We inserted a *Source*, a *Station*, a *Conveyor*, and a *Drain*. We used them with their default settings.
- Insert a *Frame* for which to define the **drag-and-drop control**. We renamed our *Frame* to *MyFrame*.

The finished simulation model looks like this:



- Define the **drag-and-drop control** for the *Frame*. Open the *Frame* and click **Edit Controls** on the **Home** ribbon tab.
- Right-click in the text box next to **drag-and-drop** and select **Create Control**.

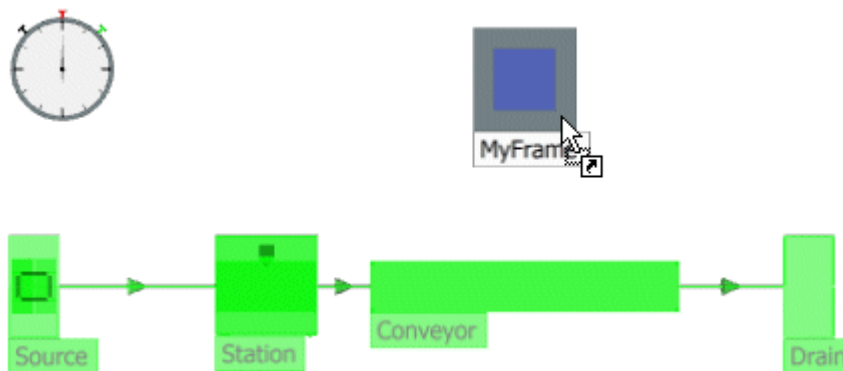


- Change the source code of the control so that it matches the instructions below:

```
param draggedObjects: object[] // one-dimensional array with n objects
                                // the size of which can change
var obj: object
debug
// @,?: drop target

for var i := 1 to draggedObjects.dim loop
  obj := draggedObjects[i]
  // enter your source code here
  print i, ": ", draggedObjects[i] // prints the dragged objects to the
  Console
next
```

- Select the production line, drag it over the *Frame*, and drop it there.



The *Console* then shows the dragged objects that were dropped.

Console

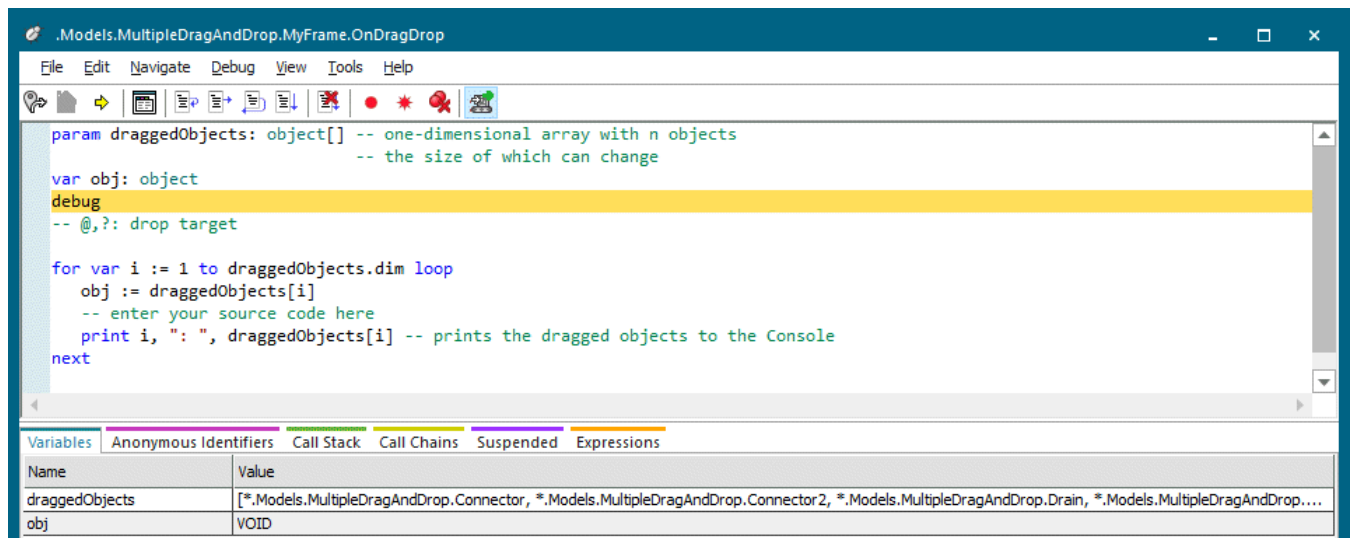
```

1: .Models.MultipleDragAndDrop.Source
2: .Models.MultipleDragAndDrop.Connector
3: .Models.MultipleDragAndDrop.Station
4: .Models.MultipleDragAndDrop.Connector2
5: .Models.MultipleDragAndDrop.Conveyor
6: .Models.MultipleDragAndDrop.Connector1
7: .Models.MultipleDragAndDrop.Drain

```

- To actually see the *array* containing the dropped objects, we added `debug` to the source code.

This opens the *Method debugger* when you drag the objects over the *Frame* and drop them there. The row with the name **draggedObjects** on the tab **Variables** then shows the *array* containing these objects.



Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.